STUDY OF ELECTRON-DOPED CUPRATES PROPERTIES:
Boltzmann transport equation and current vertex corrections

Research internship report
Non-confidentiel

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Abstract

Ce travail s'inscrit dans la continuité d'un projet visant à proposer des simulations de résistivité des cuprates dans le cadre de la théorie du transport de Boltzmann. Les cuprates sont le sujet de nombreuses études en physique de la matière condensée, cependant la physique contenue dans les mesures de transport reste encore à appréhender. Ce rapport se concentre sur la modélisation des propriétés de transport des cuprates dopés négativement.

Des mesures de résistivité sur LCCO, cuprate dopé n, ont récemment été analysées semi-quantativement avec succès dans un cadre boltzmanien, en maintenant le modèle des liquides de Fermi cohérent. La notion de *hot spots* dans le taux de collisions a permis d'obtenir des simulations de magnétorésistance en accord avec les résultats expérimentaux. L'objectif de ce projet est de proposer une analyse quantitative de la magnétoresistance et de l'effet Hall dans la même direction, en introduisant la notion de correction de vecteur de courants dans les simulations.

Nous étudions donc dans ce travail la magnétorésistance et l'effet Hall de LCCO, ainsi que l'effet des corrections de vecteur de courant sur ces grandeurs. Ces corrections sont la transcription des fluctuations de spin dans le calcul de la résistivité dans le cadre du formalisme Boltzmanien. Nous réalisons aussi dans ce rapport une brève analyse du même modèle sur le cuprate dopé p LSCO.

This work continues a project aiming at modeling numerically cuprate properties within Boltzmann transport theory's framework. Cuprates are throughoutly studied in condensed matter physics, however the physics contained in these materials remains yet to be understood. This work focuses on modelisation of transport properties in electron-doped cuprates.

Resistivity measurements in LCCO, n-doped cuprate, have recently been semi-quantitatively succesfully analysed in the Boltzmann picture and the Fermi liquid theory. The "hot spot" scattering model has allowed previous work to do magnetoresistance simulations consistent with experimental results. The present work's aim is to carry out a quantitative study of magnetoresistance and Hall effect while implementing current vertex corrections in the model.

We then study in this work LCCO longitudinal magnetoresistance and Hall effect, and the effect of current vertex corrections on these quantities. These corrections come from spin fluctuations and take part in the computations of resistivity in the Boltzmann formalism. We also briefly study the consistency of the model with hole-doped cuprates.

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Introduction

Electrons conduction governs many transport and thermodynamic properties of materials. Electrical transport, heat transport and behaviour in field follow the same core theory. This work mainly focuses on anomalous transport properties of cuprates.

In so-called conventional superconductors, superconductivity is well described by the Bardeen-Cooper-Schrieffer (BCS) theory. At very low temperature, interactions between phonons and electrons bring electrons to form so-called Cooper pairs, *i.e.* bring the metal in a superconductive state. This also is the interaction that scatters electrons in the normal state. In these materials, T_C is governed by the strength of the coupling between electrons and phonons λ_{ph}. The T-dependance of the resistivity $\rho(T)$ is tied to the scattering rate $\tau(-1)(T)$ with $\rho \propto \tau(T)$.

However, many superconductors such as heavy fermions, nickelates or cuprates, are suspected to have a non-phonon pairing mechanism. In cuprates for example, their proximity to magnetic order motivates models based on coupling to spin fluctuations [1].

The study of transport in non-superconducting states is thus the key for understanding the properties of the unconventional superconductivity in these materials. The resistivity is here dependant on a number of factors and studies have up to date mostly failed to isolate the form and strength of the scattering rate $\tau^{-1}(\varphi, T)$ from the measurements of the resistivity. Despite alternative proposals to tie the T-linear resistivity and the scattering rate, identification of the primary pairing mechanism in cuprates remains elusive [2,3].

This work focuses on cuprates, material consisting of a stack of 2D conducting CuO_2 conducting planes separated by charge reservoir planes. Outside the conducting planes, the electrons on the ions are located in tightly bound localized states situated well below the Fermi energy, essentially serving only as dopants. Therefore, the physics of transport in cuprates is mainly contained in the CuO_2 planes. Each copper atom in the conducting planes has a valence of +2 and possesses 9 electrons in the $3d$ orbital, equivalent to a single hole. The hybridization of this orbital with the O $2p$ orbital gives rise to a half-filled antibonding band at the Fermi level. The hole becomes highly localized on the Cu atom because of its strong onsite Coulomb interaction, resulting in a significant energy barrier for hopping between adjacent sites. This scenario leads to the formation of a "Mott insulator" [4].

These Mott-insulators can become metals if the material is "doped", *i.e.* if some of the interplane ions are replaced with similar in size atoms with a different valence. If the doping atom has more electrons (holes) on its valence band than the initial interplane ion, the material is said to be electron (holes) doped. Here we study primarely two doped materials derived from La_2CuO_4: the electron-doped material $La_{2-x}Ce_xCuO_4$ (LCCO) and hole-doped $La_{2-x}Sr_xCuO_4$ (LSCO).

The phase diagram of these two cuprates is well known (Figure 1), however most of the different regimes resist to any robust comprehensive theory. The study of non-superconducting regimes such as the pseudogap or the strange-metal regime are the key to solve the physics of superconductors. The antisymmetry between the electron-doped and the hole-doped parts of the phase diagram has led to think that different mechanisms may be at play to give rise to superconductivity in hole-doped cuprates than electron-doped cuprates [5].

In electron-doped cuprates, the superconducting regime arises from the antiferromagnetic regime at low doping. The mechanism of superconductivity is believed to be strongly tied to spin physics in electron-doped cuprates [5]. In this work role, we study the role of spin fluctuations and antiferromagnetic order through transport simulations. First of all, we set out the theoritical framework in which this work is situated. Then we present our methods to compute transport properties of cuprates. We detail our results on magnetoresistance and Hall effect simulations, as well as a doping and a temperature-dependance study. We briefly go through the charge-fluctuation model and a study of hole-doped cuprates. Finally, we discuss the physical relevance of our parametrisation and computations.

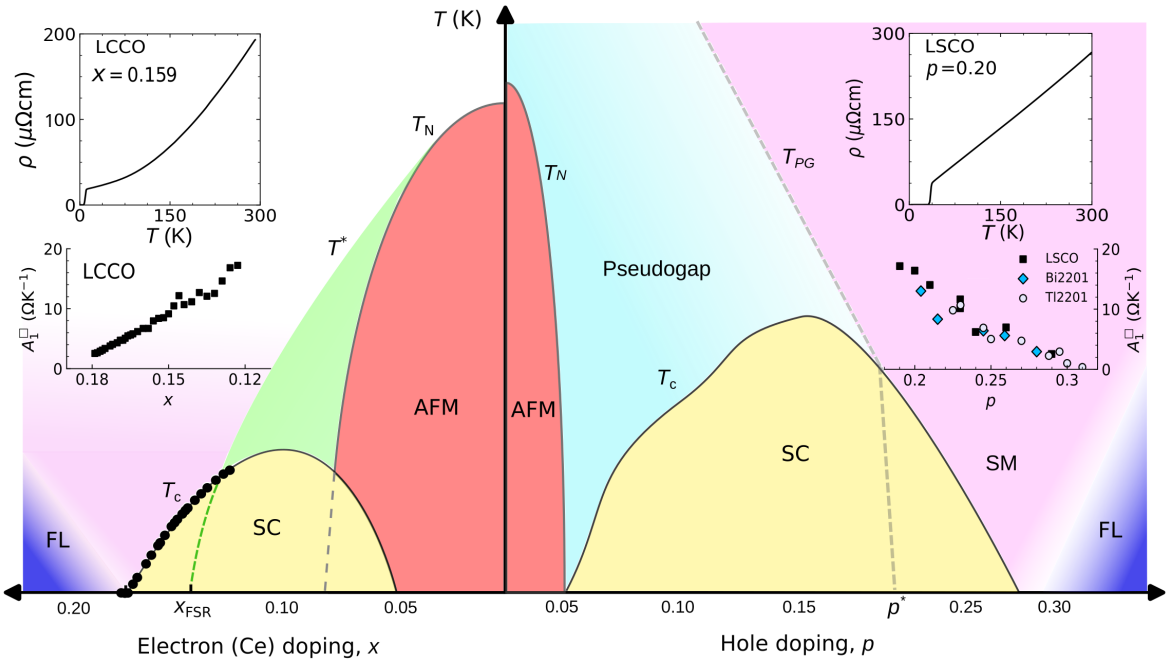


Figure 1 — Phase diagram of the electron (n)- and hole (p)- doped cuprates. Either side of the parent Mott state, the phase diagram of the cuprates exhibits a marked asymmetry. On the p-doped side, longrange antiferromagnetic (AFM) order is suppressed prior to the onset of superconductivity (SC) while on the n-doped side, AFM order coexists with SC albeit over a narrow doping range. Moreover, short range AFM correlations persist into the overdoped side (green dome bounded by T^*), causing a Fermi surface reconstruction that terminates at x_{FSR}. The p-doped cuprates host SC over a more extended range of doping that peaks at a higher maximum transition temperature. They also play host to the pseudogap phase marked in blue, bounded by T_{PG}. The strange metal (SM) phase lies beyond the pseudogap phase and is characterised by a predominantly linear-in-T resistivity with a scattering rate corresponding to the Planckian dissipation limit, an example of which is shown in the inset for LSCO at $p = 0.20$. By contrast, the resistivity of the overdoped n-doped cuprates ($x > x_{\mathrm{FSR}}$) is approximately T-linear only at the lowest temperatures, becoming quadratic above $T \approx 70$ K. The resistivity of LCCO at $x = 0.159$ is shown in the upper inset on the left. On both sides of the phase diagram, the coefficient of the T-linear resistivity $A1$ decreases in tandem with T_C, as shown in the lower insets for LCCO. Both cuprates host a correlated Fermi liquid (FL) phase in the very overdoped, non-superconducting regime. Adapted from [5]

0.1 Classical approach : Drude model

In order to fully understand the extent of the properties of cuprates, we have to first take a look at normal metals transport theory.

A first approach to the physics of transport in metals is to describe electrons as classical compact particles evolving as a gaz, *i.e.* without interacting with each other. The Drude model describes conduction electrons in metals as a classical gas of free particles with an additional fluid friction force preventing the electrons to be indefinitely accelerated.

0.1.1 Electric field

Without electric field, the electrons have velocities $\mathbf{v}_i$ randomly distributed in space, of mean value $\langle \mathbf{v}_i \rangle = 0$. However, in the presence of an electric field, the dynamic of an electron of mass m and charge $-e$ is described by:

$$m\frac{d\mathbf{v}_i}{dt} = -e\mathbf{E} \tag{1}$$

It follows that the electron gain a drag velocity $\mathbf{v} = \mathbf{v}_{i(0)} - et/m\mathbf{E}$, which tends towards infinity with time.

In a classical electron gas, electrons can only travel freely a finite time before they undergo a collision. These collisions randomise the velocity, averaging it out and enabling it to stay finite. Such collisions are random and happen with probability dt/τ during time dt. The fraction dt/τ that has been collided has no contribution to the mean value of the velocity. The fraction $(1 - dt/\tau)$ is being accelerated with a new velocity $\mathbf{v}_i(t + dt) = \mathbf{v}_i(t) - 1/3\, e\mathbf{E}dt$. The total contribution to the mean velocity is then at first order:

$$\langle \mathbf{v}_i(t + dt) \rangle = \langle \mathbf{v}_i(t) \rangle - e\frac{dt}{m}\mathbf{E} \tag{2}$$

Which gives in the steady state an equal contribution between the electrostatic force and the fluid friction force related to collisions happening at the scattering rate τ and proportional to $\mathbf{v}$:

$$\mathbf{0} = -e\mathbf{E} - m\frac{\langle \mathbf{v}_i \rangle}{\tau} \tag{3}$$

We can then compute the current density $\boldsymbol{j}$ $\boldsymbol{j} = -e\sum_i \mathbf{v}_i = -ne\langle \mathbf{v}_i \rangle$ This current density then becomes :

$$\boldsymbol{j} = \frac{ne^2\tau}{m}\mathbf{E} \tag{4}$$

With Ohm's law $\boldsymbol{j} = \sigma\mathbf{E}$ we get the conductivity and the resistivity ρ :

$$\sigma = \frac{ne^2\tau}{m} := \frac{1}{\rho} \tag{5}$$

0.1.2 Effect of a magnetic field

When applying an external and transverse magnetic field $\mathbf{B} = B\boldsymbol{e}_z$ the Lorentz force $\mathbf{v} \times \mathbf{B}$ makes the electrons devate from their straight trajectory to a cyclotron motion. The motion equation becomes in the steady-state regime:

$$\mathbf{0} = -e(\mathbf{E} + \langle \mathbf{v}_i \rangle \times \mathbf{B}) - m\frac{\langle \mathbf{v}_i \rangle}{\tau} \tag{6}$$

In general the current density is thus no longer perpendicular to the electric field and the conductivity becomes thus a tensor:

$$\boldsymbol{J} = \sum_{ij} \sigma_{ij} E_j \boldsymbol{e}_i \qquad \{i,j\} \in \{x,y\} \tag{7}$$

We can compute it within the steady state regime :

$$\sigma_{ij} = \frac{ne^2\tau}{m} \begin{pmatrix} 1 & -\frac{m}{eB\tau} \\ \frac{m}{eB\tau} & 1 \end{pmatrix} \tag{8}$$

Experimentally, the resistivity is measured with a standard geometry. B is perpendicular to the material's plane $\mathbf{B} = B\boldsymbol{e_z}$. The electric field is applied along the x direction. We call the resistivity in the (x, y) plane the longitudinal ρ_{xx} for the resistivity measured along the direction of the electric field, and the transverse resistivity (or Hall resistivity) for the resistivity ρ_{xy} measured along y. We can compute the resistivity $\rho_{ij} \coloneqq \sigma_{ij}^{-1}$ so that $\mathbf{E} = \sum_{ij} \rho_{ij} j_j \boldsymbol{e}_i$ We have

$$\rho_{ij} = \frac{1}{\det(\sigma_{ij})} \begin{pmatrix} \sigma_{yy} & -\sigma_{xy} \\ -\sigma_{yx} & \sigma_{xx} \end{pmatrix} \tag{9}$$

Which gives

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2} = \rho_{xx} = \frac{1}{\sigma(B=0)} \qquad \text{and} \qquad \rho_{xy} = -\frac{\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2} = \rho_{xy} = \frac{B}{ne} \tag{10}$$

The Drude model is a simple but powerful approach to the physics of solids. However, it still reaches its limits for fundamental properties of metals. For example, the Drude model predicts no dependance of the longitudinal resistivity, *i.e.* no magnetoresistance (MR). It also predicts a constant Hall number $R_H \coloneqq \rho_{xy}/B$ with no dependance either in field or temperature. In this work we focus on anomalous propeties of metals, *i.e.* anomalous behaviour of the MR, the Hall effect, etc. We need a more detailed framework in order to have a more accurate description of the solid.

0.2 Quantum description of a solid

Condensed matter physics revolves around the quantum description of solids. If we look at the Hamiltonian of a such a system, we have to take into account the electrical interaction between nuclei, between electrons and nuclei, and finally between electrons. This Hamiltonian may be exact, it is also unsolvable, we must thus adopt different approximations and ways of modelling the system. A first way to describe the solid is to consider that ions constituing the lattice nodes have a fixed spatial position but the electrons of the solid are treated as quantum wavevectors.

Electrons are fermions, their dynamic obeys thus Pauli's exclusion principle, which statues that two electrons cannot be in the same state. Pauli's principle combined with the geometry of the lattice leads to the formation of bands of allowed states that the electrons can occupy.

In a crystal, electrons evolve in a periodic potential, created by the positively charged ions constituing the metal. Their wave function follows Bloch's theorem :

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n,\mathbf{k}}(\mathbf{r}) \tag{11}$$

where $u_{n,\mathbf{k}}(\mathbf{r})$ is a periodic function for the lattice vectors in real space. If the $\mathbf{k}$-vector is shifted by a reciprocical lattice vector $\mathbf{G}$, $\psi_{n,\mathbf{k}+\mathbf{G}}(\mathbf{r})$ is also a solution of Schrödinger's equation for the same energy. The symmetry of the lattice maps all states back into the first Brillouin zone (BZ), that is the unitary cell in $\mathbf{k}$-space.

The wave vector $\mathbf{k}$ in the BZ can be associated with the electron's momentum $\mathbf{p} = \hbar\mathbf{k}$. We can also consider that electrons have a defined momentum as long as the wave packet spread in reciprocical space is smaller compared to the size of the BZ, *i.e* $\Delta\mathbf{k} \ll \text{BZ}$ or equivalently $\Delta\mathbf{r} \gg a$ where a is a lattice parameter in real space.

We can define the band velocity of the electron, which is no other than the group velocity of the wavepacket, normal to the surface of constant energy :

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar}\nabla_{\mathbf{k}}\varepsilon_n(\mathbf{k}) \tag{12}$$

Under a external force $\mathbf{F}$, the energy of the electron is modified, so is the wave vector. We have :

$$\begin{aligned} \mathrm{d}W = \mathrm{d}\varepsilon_n(\mathbf{k}) &= \mathbf{F}\cdot\mathbf{v}\,\mathrm{d}t \\ &= \mathbf{F}\cdot\frac{1}{\hbar}\nabla_{\mathbf{k}}\varepsilon_n(\mathbf{k}) = \nabla_{\mathbf{k}}\varepsilon_n(\mathbf{k})\cdot\mathrm{d}\mathbf{k} \end{aligned} \tag{13}$$

Thus we obtain an equation that strenghens $\hbar\mathbf{k}$ as the momentum of the electron

$$\hbar\frac{\mathrm{d}\mathbf{k}}{\mathrm{d}t} = \mathbf{F} \tag{14}$$

Differenciating the velocity with respect to time, we get with the chain's rule :

$$\frac{\mathrm{d}v_i}{\mathrm{d}t} = \sum_{j=x,y,z}\frac{\partial v_i}{\partial k_j}\frac{\partial k_j}{\partial t} = \frac{1}{\hbar^2}\sum_j\frac{\partial^2\varepsilon}{\partial k_j\partial k_i}F_j \tag{15}$$

With this Newton-like equation, we can define the effective mass of an energy band as the tensor

$$\left(\frac{1}{m_n^*}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2\varepsilon_{n(\mathbf{k})}}{\partial k_i\partial k_j} \tag{16}$$

Close to the extrema, the wave packet is almost quadratic, the band mass is locally constant in $\mathbf{k}$.

All the previous treatment is valid for one electron in one otherwise band. We also want to look at what happens when we consider the physics of a "missing" electron in one filled band, *i.e.* a hole. The current in a band is the sum of every filled state contribution $-ne\mathbf{v} = -e\mathbf{v}(\mathbf{k})/L$. The current of a completely filled band is thus in case of a symmetrical FS (which is our case in this work) :

$$\mathbf{I} := 2\sum_{\mathbf{k}} -e\frac{\mathbf{v}(\mathbf{k})}{L} = -2\frac{e}{L}\sum_{\mathbf{k}}\frac{1}{\hbar}\nabla_{\mathbf{k}}\varepsilon(\mathbf{k}) = 0 \tag{17}$$

If follows that if we look at a so-called hole, its contribution to the current is then $\mathbf{I}_h = +e\mathbf{v}(\mathbf{k})/L$. A hole and an electron have thus opposite charges $-e$ and $+e$.

The Fermi liquid theory, developed by Landau in 1956 is based on the adiabatic continuity [6]. It enables us to treat the interactions of electrons in a metal with only slight modifications in our model. Starting from an electron gas with no interaction, as we "turn on" the interaction's intensity adiabatically, non-interacting states should evolve smoothly into interacting states. In this case, intrinsic properties of the solid are conserved: quantum numbers stay relevant to the system, electric charge, spin and momentum are conserved. However, dynamical properties such as the band mass, magnetic moments, etc. are renormalised due to the interactions. It is said that the electrons are "dressed" with interactions.

A basic definition of a metal is a material whose resistivity increases with temperature. As temperature increases, the scattering rate τ^{-1} increases too *i.e.* the lifetime of quasiparticles decreases,

which leads to the resistivity to increase. We can define the mean free path as $\ell(T) = v_F\tau$ with v_F the Fermi velocity. Thus as the resistivity increases, the mean free path decreases. However, the metallic state cannot persist as ℓ drops to 0 infinitely. For the model to stay consistent, ℓ has to be greater than the lattice spacing a. As the mean free path reaches its limit, the resistivity saturates, this is called the Motte-Ioffe-Regel limit (MIR) [7].

In strongly correlated electron systems, we measure a non-saturated T-linear resistivity up to extremely high T, beyond the MIR limit. Such metal are referred as "bad" metals. High T_C superconductors, here cuprates, are bad metals, and we will try in this work to develop a model within the Boltzmann transport picture to fit their resistivity behaviour.

0.2.1 Boltzmann transport equations

The Boltzmann transport equation (BTE) is a theoretical framework used to describe metals. Instead of studying the dynamical properties of electrons (position, momentum) and adding their individual contributions, we relocate the problem in terms of distribution function $f(\mathbf{r}, \mathbf{k}, t)$. The description of the local concentration of electrons ("occupancy" of states) and its evolution encodes fully the system.

We will establish the BTE using a semi-classical approach of the dynamic of the system (kinetic theory of gases).

At thermodynamic equilibrium, the distribution of the occupation of states is given by Fermi's distribution:

$$f_0(\varepsilon) = \frac{1}{1 + e^{\beta(\varepsilon-\mu)}} \quad \text{where} \quad \beta = \frac{1}{k_B T} \tag{18}$$

Out of equilibrium, we define the distribution as $f(\mathbf{r}, \mathbf{k}, t)$, which can be both time dependant and position dependant. At $T = 0$, the distribution function f_0 is 1 for $\varepsilon < \mu$ and 0 for $\varepsilon > \mu$, *i.e.* all states are occupied up to $\varepsilon = \mu$, this energy is called the Fermi energy ε_F. We call the Fermi Surface (FS) the surface corresponding to the isonergy $\varepsilon = \varepsilon_F$.

The thermodynamic equilibrium state is simply the fundamental where all electronic states of lower energy are occupied, taking into account Pauli's principle. If the Fermi level crosses an energy band, electrons can jump from one occupied state to an empty state easily: an electric field can induce a current, the solid is a metal. However, if the Fermi level is in a gap between two bands, empty states become inaccessible for electrons, the solid is an insulator.

This distribution enables us to compute macroscopic quantities of the system. With the volume element in phase space being $\mathrm{d}\mathbf{k} \cdot \mathrm{d}\mathbf{r} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}k_x\,\mathrm{d}k_y\,\mathrm{d}k_z$, taking into account the spin-degeneracy we can explicit the mean number of particles as :

$$N(t) = 2\int \mathrm{d}^3\mathbf{r} \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} f(\mathbf{r}, \mathbf{k}, t) \tag{19}$$

For an homogeneous system at equilibrium and without any external perturbation, we can more generally the average of any physical value A as

$$\langle A \rangle = \frac{\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} 2A(\mathbf{k}) f_0(\mathbf{k})}{\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} 2f(\mathbf{k})} = \frac{1}{n}\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} 2A(\mathbf{k}) f(\mathbf{k}) \tag{20}$$

The average velocity is then expressed as :

$$\langle v_i \rangle = \frac{1}{n} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} 2 v_i(\mathbf{k}) f(\mathbf{k}) \tag{21}$$

Especially we have with the equilibrium distribution f_0:

$$\langle v_i \rangle = \frac{1}{n} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} 2 v_i(\mathbf{k}) f_0(\mathbf{k}) = 0 \tag{22}$$

as $\mathbf{v}(\mathbf{k})$ is an odd function and f_0 is an even function. As expected there is no contribution of the equilibrium distribution to the average velocity.

To compute the current density of electrons we have:

$$J_i = (-e) n \langle v_i \rangle \tag{23}$$

As seen in eq. 22, only the out-of-equilibrium distriubtion $f - f_0$ contributes to the average of the velocity. We can then write the current density vector as

$$J_i = -\frac{2e}{(2\pi)^3} \int \mathrm{d}^3 k \, (f - f_0) v_i \tag{24}$$

The Fermi distribution is brought out of equilibrium by external fields. This introduces a time evolution to f. As the net current flow is 0, we have for the equilibrium distribution $\mathrm{d}f_0/\mathrm{d}t = 0$. Out of equilibrium, this quantity is finite. Transfer between states encodes a *collision integral* $I[f]$ that works to relax the system back to equilibrium. This integral encodes the details of relaxing and scattering processes that allow electrons to change states. We have

$$\frac{\mathrm{d}f}{\mathrm{d}t} = I[f] \tag{25}$$

This integral can be in principle extremely complex. A simple example can be computed through Fermi's golden rule, *i.e.* considering elastic scattering only. The scattering probability between two states $\mathbf{k}$ and $\mathbf{k}'$ is $\frac{1}{\Omega}\omega(\mathbf{k}, \mathbf{k}')\delta(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})$, with $\omega(\mathbf{k}, \mathbf{k}')$ the probability of state $\mathbf{k}'$ to scatter to state $\mathbf{k}$. The transfer integral is then one if and only if $\varepsilon_{\text{initial}} = \varepsilon_{\text{final}}$. We have in this case :

$$I[f] = -\frac{1}{\Omega} \sum_{\mathbf{k}'} \left[\omega(\mathbf{k}, \mathbf{k}') f_{\mathbf{k}} (1 - f_{\mathbf{k}'}) - \omega(\mathbf{k}', \mathbf{k}) f_{\mathbf{k}'} (1 - f_{\mathbf{k}}) \right] \delta(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) \tag{26}$$

The time evolution of the distribution function can be rewritten with the chain rule as:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \left(\frac{\partial}{\partial t} + \frac{\partial \mathbf{r}}{\partial t} \cdot \boldsymbol{\nabla}_{\mathbf{r}} + \frac{\partial \mathbf{p}}{\partial t} \cdot \boldsymbol{\nabla}_{\mathbf{p}} \right) f(\mathbf{r}, \mathbf{p}, t) \tag{27}$$

This can be rewritten as

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \overbrace{\partial_t}^{\text{temporal term}} + \overbrace{\mathrm{v} \cdot \boldsymbol{\nabla}_{\mathbf{r}}}^{\text{diffusion term}} + \overbrace{\mathrm{F} \cdot \boldsymbol{\nabla}_{\mathbf{p}}}^{\text{Force term}}) f(\mathbf{r}, \mathbf{p}, t) = I[f] \tag{28}$$

The temporal term encodes any explicit time dependance of the distribution function in a non-static steady state, caused for example by time dependant fields, AC conductivity, optical properties, etc. The diffusion term encodes the diffusion motion out of regions of real space. This can be spatial variation in fields such as temperature gradients. The force term encodes the influence of field on the distribution function through momentum (static fields, conductivity, etc).

We often work in the Boltzmann transport picture with the relaxation time approximation (RTA) *i.e.* with a linear scattering acting as an effective linear restoring "force".

$$I[f] = \frac{f - f_0}{\tau} \tag{29}$$

Defining the difference from equilibrium $g(t)$ as $g(t) := f - f_0$ we have the following equation within the RTA:

$$I[f] = \frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\mathrm{d}f_0 + g(t)}{\mathrm{d}t} = \frac{\mathrm{d}g}{\mathrm{d}t} = -\frac{1}{\tau}g(t) \tag{30}$$

The solution of $g(t)$ is then simply

$$\begin{aligned} g(t) &= g(0)\,\mathrm{e}^{-t/\tau} \\ f(t) &= f_0 + g(0)\,\mathrm{e}^{-t/\tau} \end{aligned} \tag{31}$$

In the RTA, the perturbations die off exponentially: the relaxation time is a decay time for restoring the equilibrium.

This basic theoritical framework does not cover anomalous effects of the high temperature superconductors. The aim of this work is to reconcile BTE and the transport behaviour of electron doped cuprates.

1 Methods

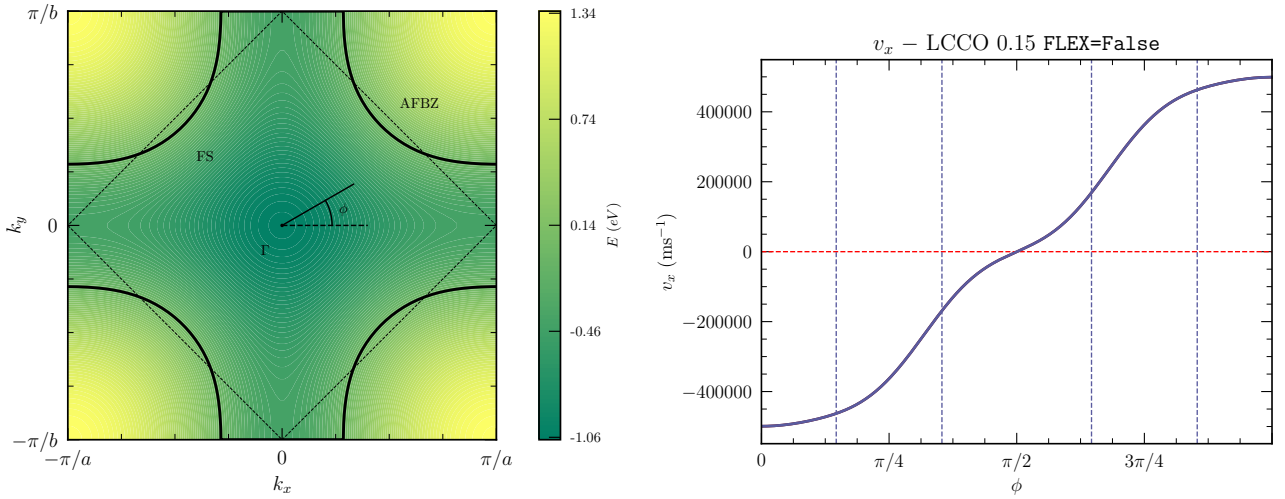
This work contains computations based on the code stucture of a previous broject done by A. Cuoghi [8]. In this project, the physical properties of the metal are computed using the tight-binding model with coefficients determined from [5,9]. The response to field and temperature variations (*i.e.* the resistivity) is computed within the Boltzmann picture using a numerical solution of the Shockley-Chambers tube integral formula. Based on these computations, this work features additional elements compared to the original simulations of A. Cuoghi [8], that is the implementation of electron-doped cuprates properties, of hotspots in the scattering rate and of current vertex corrections.

1.1 Energy dispersion and Fermi surface

All simulations are done on virtual samples. LCCO samples [5] as well as hole-doped samples [8] are numerically representated by real space lattice vectors a, b and c, chemical potential μ and tight binding parameters. The undoped compound of the cuprates we study is a Mott insulator. Due to strong electron repulsion, the electrons are strongly localized on Cu atoms. This motivates the use of tight-binding parametrisation. The energy dispersion relation in the (k_x, k_y) plane is computed using

$$\begin{aligned}\varepsilon(k_x,k_y) = \mu &+ \frac{t_1}{2}\left[\cos(k_x a) + \cos(k_y b)\right] \\ &+ t_2 \cos(k_x a)\cos(k_y b) \\ &+ \frac{t_3}{2}\left[\cos(2k_x a) + \cos(2k_y b)\right] \\ &+ \frac{t_4}{2}\left[\cos(2k_x a)\cos(k_y b) + \cos(2k_x b)\cos(k_y a)\right]\end{aligned} \tag{32}$$

The Fermi surface is described by the equation $\varepsilon_{\mathbf{k}} = 0$ and the velocity is computed using $\mathbf{v}_{\mathbf{k}} = \frac{1}{\hbar}\boldsymbol{\nabla}_{\mathbf{k}}\varepsilon$. As an example, Fermi surface and v_x for LCCO are shown in Figure 2



(a) (b)

Figure 2 — (a) Fermi surface of LCCO $x = 0.15$ computed with tight-binding, it is purely hole-like. To compute kinetic quantities in the rest of this work, we shift the FS by (π, π) and center it around the M point. Dotted line represent the antiferromagnetic Brillouin zone (AFBZ) (b) Fermi velocity computed with $\hbar\mathbf{v}_\mathbf{k} = \boldsymbol{\nabla}_\mathbf{k}\varepsilon$, along k_x axis for LCCO $x = 0.15$

1.2 Shockley-Chambers tube integral formalism

The SCTIF is a convenient and accurate formula to compute the resistivity suitable for high magnetic fields and 2D systems (both p-doped and n-doped cuprates are quasi 2D materials). The starting point is the Boltzmann transport equation for a uniform systeme under a magnetic field, within the RTA.

$$\frac{\partial f}{\partial t} + \frac{\partial f}{\partial \mathbf{k}} \cdot \frac{-e}{\hbar}(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = -\frac{f - f_0}{\tau} \tag{33}$$

where f_0 is the Fermi distribution at thermodynamical equilibrum and f is the Fermi distribution at time t and temperature T. We define the displacement from equilibrium as $f_1 = f - f_0$. Knowing that $\boldsymbol{\nabla}_\mathbf{k}\varepsilon = \hbar\mathbf{v}$, we have

$$\frac{\partial}{\partial t}\left(\mathrm{e}^{t/\tau} f_1\right) = e\frac{\partial f}{\partial \varepsilon}\mathrm{e}^{t/\tau}\mathbf{v}(t) \cdot \mathbf{E} \tag{34}$$

whose solution is given by

$$f_1 = e\int_{-\infty}^{0} \mathrm{d}t \frac{\partial f}{\partial \varepsilon}\mathrm{e}^{t/\tau}\mathbf{v}(t) \cdot \mathbf{E} \tag{35}$$

We compute the conductivity using the expression of the current density in eq. 24 and the expression of the conductivity tensor eq. 7 :

$$\sigma_{ij} = -\frac{e^2}{4\pi^3} \int \mathrm{d}^3k \, v_i(0) \int_{-\infty}^{0} \mathrm{d}t \, \frac{\partial f}{\partial \varepsilon} \, v_j(t) \, \exp\left(\int_t^0 \frac{\mathrm{d}t'}{\tau(t')} \right) \tag{36}$$

This expression is the Shockley-Chanbers tube-integral formula (SCTIF) for the conductivity [10,11]. The effect of the magnetic field is taken into account in the computation of the velocity in the first time integral through the cyclotron motion. The strength of this formula is exact for several orders in $\mathbf{B}$ while being numerically computable in reasonnable time.

To compute this expression we can do further simplications. First of all, in cuprates, the term of thermal broadening is usually negligible ($E_F \gg k_B T$), we can thus approximate $\partial f / \partial \varepsilon$ with a delta function. The integral over all of k-space becomes an integral over the Fermi surface only. We can also assume a cylindric shape for the Fermi surface, the integral becomes effectively one of a tube. This is then possible to integrate over the $2D$ FS, taking into account the z expansion of the FS as a factor $2\pi/c$

$$\sigma_{ij} = -\frac{e^2}{2\pi^2 c\hbar} \oint_{\mathrm{FS}} \mathrm{d}k_t \, \frac{v_i(0)}{v_F} \int_0^{+\infty} \mathrm{d}t \, v_j(-t) \, \exp\left(-\int_0^t \frac{\mathrm{d}t'}{\tau(-t')} \right) \tag{37}$$

This expression is used to compute the resistivity with simply inversing the tensor. Several numerical methods are to be noted :

- The time between 0 and $+\infty$ is in our simulations a great finite amount of time, typically 100 times bigger than the maximal average lifetime of electrons (*i.e.* the inverse of the minimum scattering rate)
- The evolution of the velocity is computed using the Runge-Kutta algorithm at fourth order. This evolution depends on the strength of the magnetic field: the stronger $\mathbf{B}$, the faster electrons travel around the FS, the more the velocity evolves in time compared to its initial value.
- Both integral in time and k-space are computed using `trapezoid` method from the python `numpy` library with $\mathbf{k}$ and $\mathbf{v}$ discrete arrays.

1.3 Current vertex corrections (CVC)

The key contribution of this study is the implementation of current vertex corrections (CVC) in the SCTIF. In interacting electron systems, an excited electron induces other particle-hole excitations by collisions. Thanks to the fluctuation-exchange (FLEX) approximation, the CVC are easily incorporated into the Boltzmann Transport equations.

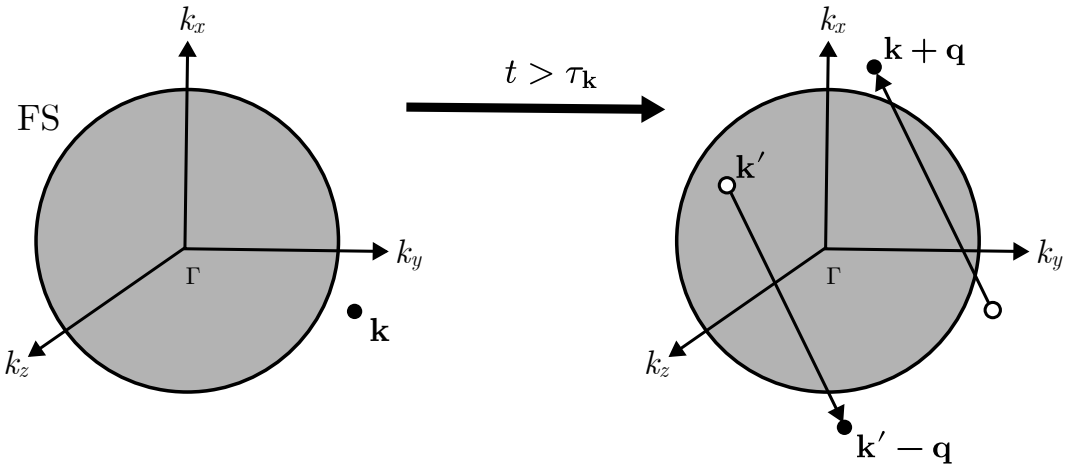


Figure 3 — Representation of the backflow process in the scattering after time $\tau_{\mathbf{k}}$. Instead of just "vanishing" the state $\mathbf{k}$ decays to $\mathbf{k}+\mathbf{q}$. A pair electron-hole is created between these two points. Another pair electron-hole is created at $k', k'+q$ in order to respect conservation laws. Adapted from [12]

In the RTA, we consider an excited quasiparticle at $\mathbf{k}$ (Figure 3), after a time $\tau_{\mathbf{k}}$, the quasiparticle decays as if it annihilates after the relaxation time. However, this unphysical process allowed by the RTA yields a violation of the conservation law $\partial\rho/\partial t + \boldsymbol{\nabla}\cdot(\mathbf{j}) = 0$ [12]. This law is a very important constraint for transport properties. In order to study anomalous transport phenomena, we have to take into account what Landau calls backflow in the phenomenological Fermi liquid theory [6], *i.e.* the current vertex corrections. The CVC represents the induced current due to these particle-hole excitations.

When a quasiparticle decays, it decays from $\mathbf{k}$ to $\mathbf{k}+\mathbf{q}$ and a particle-hole pair should be created between $\mathbf{k}'$ and $\mathbf{k}'-\mathbf{q}$ The CVC is thus closely related to the momentum conservation law.

Cuprates are doped Mott insultors, *i.e.* nearly antiferromagnetic (AFM) metals. In these metals, there is a strong backwards scattering due to antiferromagnetic fluctuations that induces CVC. A quasiparticle at $\mathbf{k}$ is scattered to $\mathbf{k}+\mathbf{q}$ with $\mathbf{q}\sim\mathbf{Q}$, where $\mathbf{Q}=(\pm\pi,\pm\pi)$ is the antiferromagnetic wavevector. A particle-hole pair is thus created at $(\mathbf{k}',\mathbf{k}'-\mathbf{q})$, however, this current is almost cancelled after summation on all possible $\mathbf{k}'$. The CVC is given by the quasiparticle $\mathbf{k}+\mathbf{Q}$ alone [12].

Transport phenomena in HTSCs based on the Fermi liquid theory can be studied using the spin fluctuation theory. To derive the formula of the current within the CVC, Kontani uses the fluctuation exchange (FLEX) approximation [12] based on the following Hamiltonian from the Hubbard model.

$$H = \sum_{\mathbf{k},\sigma} \varepsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k},\sigma} c_{\mathbf{k}\sigma} + U \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}} c^{\dagger}_{\mathbf{k}+\mathbf{q},\uparrow} c^{\dagger}_{\mathbf{k}'-\mathbf{q},\downarrow} c_{\mathbf{k},\downarrow} c_{\mathbf{k}',\downarrow} \tag{38}$$

where U is the Coulomb on-site Coulomb repulsion, $\varepsilon_{\mathbf{k}}$ is the spectrum of the conduction electron, and $c^{\dagger}_{\mathbf{k},\sigma}$ is the creation operator of an electron with momentum $\mathbf{k}$ and spin σ.

The FLEX approximation is suitable for the analysis of nearly AF Fermi liquids [12]. Treating these materials with strong AF fluctuations introduces the concept of hot spots of the quasiparticule damping rate. That is, $\tau_{\mathbf{k}}^{-1}$ becomes anisotropic in the presence of AF fluctuation. The points of the FS where $\tau_{\mathbf{k}}^{-1}$ takes a maximum are called hot spots, and cold spots where it takes a minimum (see Figure 4a). Hot spots usually exist abround the crossing points with the AF Brillouin zone (AFBZ)-boundary and the Brillouin zone boundary, *i.e.* where the $\mathbf{Q}$ vector links two existing points of the Brillouin zone (Figure 4b). Cold spots are at the points where the distance from the AFBZ-boundary is the largest.

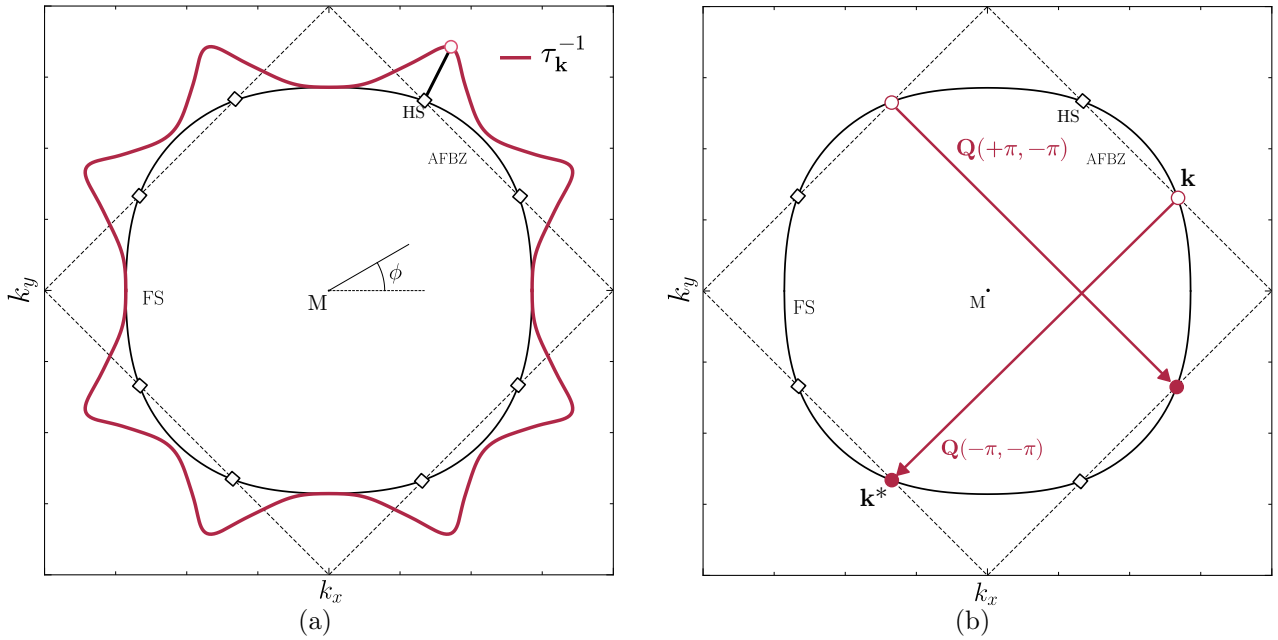


Figure 4 — (a) The scattering rate $\tau_{\mathbf{k}}^{-1}$ takes its maximum at the intersection between the AFBZ boundary and the FS. This points are called hot spots. (b) Representation of the scattering process by the vector $Q = (\pm\pi, \pm\pi)$. The state $\mathbf{k}$ is mixed with the state $\mathbf{k}^*$ at $\mathbf{k} + \mathbf{Q}$, inducing a modification in form of the current

In nearly AFM metals, under the FLEX approximation, the total current $\boldsymbol{J}_{\mathbf{k}}$ can be computed written as:

$$\boldsymbol{J}(\mathbf{k}) = \frac{1}{1 - \varepsilon_{\mathbf{k}}^2} \left(\mathbf{v}_{\mathbf{k}} + \varepsilon_{\mathbf{k}} \mathbf{v}_{\mathbf{k}+\mathbf{Q}} \right), \tag{39}$$

where $Q = (\pm\pi, \pm\pi)$ mixes two states due to spin fluctuations, $\mathbf{v}_{\mathbf{k}}$ is the Femi velocity at point $\mathbf{k}$ and $\varepsilon_{\mathbf{k}}$ is a factor between 0 at the cold spots and 1 at the hot spots and represents the strength of the AFM correlations. Typically, $\varepsilon_{\mathbf{k}} \approx 1 - c/\xi_{\mathrm{AFM}}^2$ where ξ_{AFM} is the antiferromagnetic correlation length and takes it maximum at the hot spots. The current $\boldsymbol{J}_{\mathbf{k}}$ is then no longer parallel to the Fermi velocity itself, thus not normal to the Fermi surface (Figure 5).

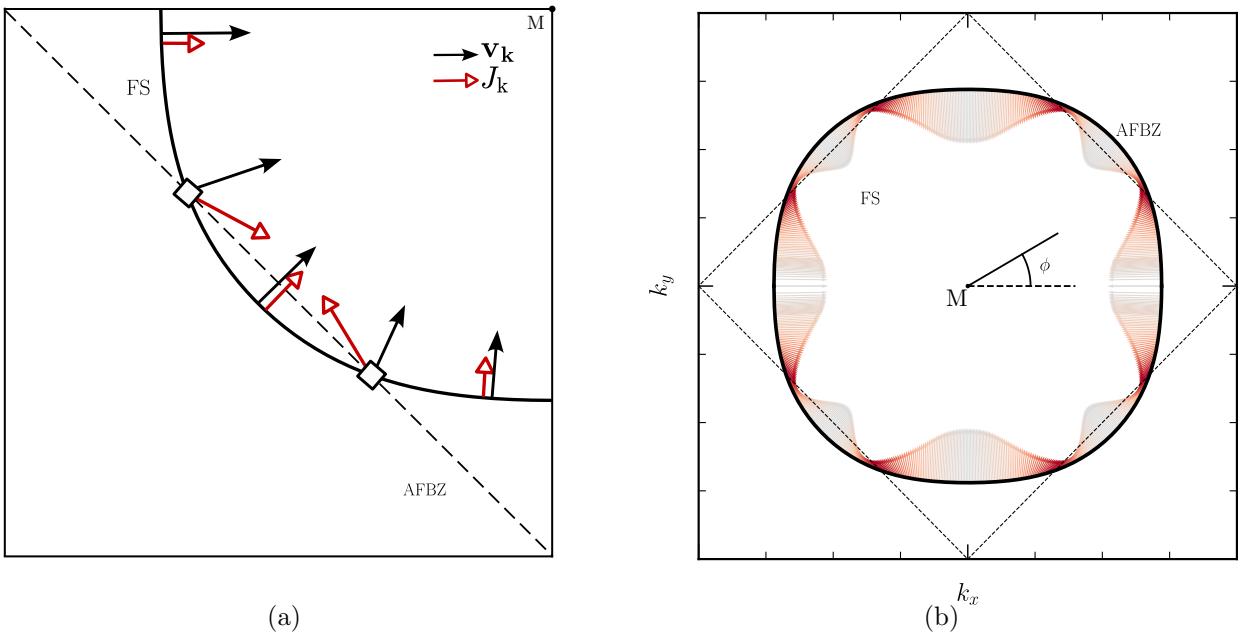


Figure 5 — (a) Schematic representation of the effect of the CVC, the current is no longer parralel to the Fermi velocity close to the hot spots. (b) Fermi surface with vectors corresponding to corrected mean free path vectors $\boldsymbol{l} = (J_x\tau, J_y\tau)$. Red regions are regions where the mean free path is no longer parrallel to the velocity. The curvature of the mean free path induces region where the circulation of ℓ changes sign

1.3.1 CVC and resistivity

We introduce the current vertex corrections in the computation of the resistivity in the Boltzmann picture in order to better reproduce anomalous transport phenomena. Experimental results show that the Hall number R_H in electron-doped cuprate LCCO decreases with temperature, and even changes sign (see Figure 6).

In order to better understand the qualitative effects of the CVC on the resistivity, we derive the Jones-Zener expansion of the Boltzmann transport equation [14]. This approximation of the Boltzmann transport equation is only applicable at low magnetic fields, however R_H itself has only been measured to date at relatively low field strengths. This form also allows the use of analytical expressions for most of the key quantities, integrated over the Fermi surface.

$$\begin{aligned} \sigma_{xx} &= \frac{e^2}{2\pi^2\hbar d}\int_0^{2\pi} \mathrm{d}\varphi\, k_F(\varphi) l_x^*(\varphi)\,\frac{\cos(\varphi-\gamma)}{\cos(\gamma)} \\ \sigma_{xy} &= -\frac{e^3}{2\pi^2\hbar^2 d}\int_0^{2\pi} \mathrm{d}\varphi\, l_x^*(\varphi)\,\frac{\partial l_y^*}{\partial\varphi} \end{aligned} \tag{40}$$

where $l_x^* = J_x/\Gamma(\varphi)$ and $\Gamma = 1/\tau$ is the mean free path modified by the current vertex corrections and $\gamma = \arctan\left(\frac{1}{k_F}\partial k_F/\partial\varphi\right)$

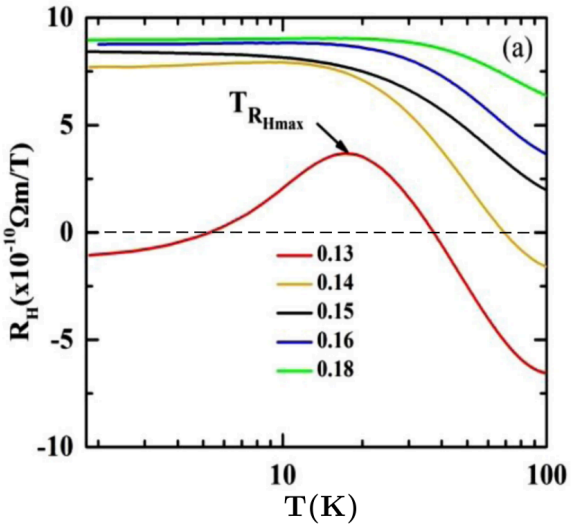


Figure 6 — Hall coefficient in LCCO $R_H(T)$ curves in non-combinatorial thin films of LCCO over the doping range $0.13 \leq x \leq 0.18$ for a field of 14T. R_H decreases with increasing T and tend towards negative values even for overdoped samples beyond 0.14 Reproduced from [13]

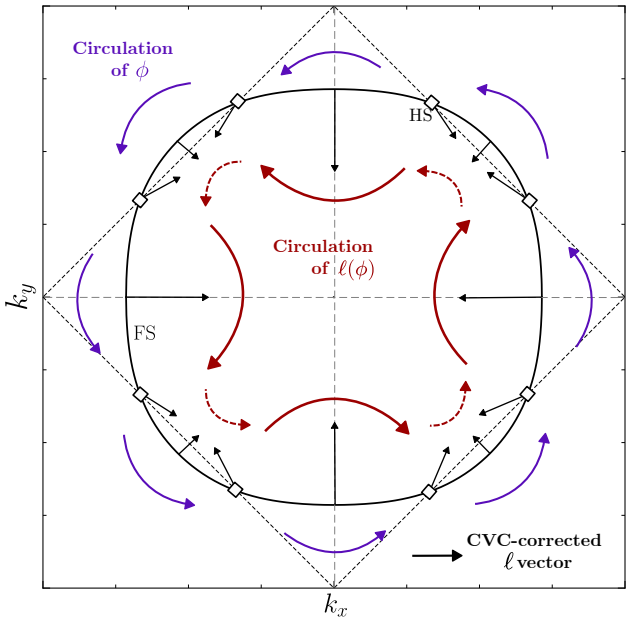


Figure 7 — Representation of the change in circulation of $\ell(\varphi)$ due to CVCs for the case where ℓ is shorter at the nodes than at the anti-nodes. The length of the black arrows reflect the magnitude of $\ell(\varphi)$. Wherever the circulation of $\ell(\varphi)$ opposes that of φ (*i.e.* is anticlockwise), the contribution to σ_{xy} from that section is negative. σ_{xy} itself is dominated by those regions of the FS where the curvature is largest. For LCCO, the largest curvature occurs around the zone diagonals. Adapted from [5]

In the absence of anisotropy in $\ell(\varphi)$ or any CVCs, $l_x\,\partial l_y/\partial\varphi$ is always positive. Introducing the CVCs, however, leads to φ-ranges where the contribution in the circulation is negative. Within the Ong construction, this sign change arises due to a change in circulation of the ℓ-vector around the FS [15]. These regions of opposite circulation can contribute a component to σ_{xy} of opposite sign, even when the scattering time τ is isotropic Figure 7. The width of these regions is set by the width of the

"influence" of the CVCs. Incorporating anisotropy into $\tau(\varphi)$ will act to enhance or reduce this effect, depending on the form of the scattering rate itself. For $R_{H(T)}$ to drop in magnitude with increasing T relative to its isotropic, low-T value, we require that ℓ in those regions in which the circulation of $\ell(\varphi)$ is opposite to that of φ to be longer [5].

The magnetoresistance is less sensible to the anisotropy in the mean free path. It is then expected that the CVC don't modify its behaviour too much compared to what was previously computed in [5]. Its magnitude in T and $\mathbf{B}$ scales with the magnitude of the scattering rate.

1.3.2 Computation of the CVC

In order to incorporate CVCs into our Boltzmann framework, we substitute the velocity in the SCTIF by $\boldsymbol{J}_{\mathbf{k}}$. We have with current vertex corrections:

$$\sigma_{ij} = -\frac{e^2}{2\pi^2 c\hbar} \oint_{\mathrm{FS}} \mathrm{d}k_t \, \frac{J_i(0)}{v_F} \int_0^{+\infty} \mathrm{d}t \, J_j(-t) \, \exp\left(- \int_0^t \frac{\mathrm{d}t'}{\tau(-t')} \right) \tag{41}$$

The time evolution of the current is still computed with the Runge-Kutta algorithm, however it is to be noted that there is no Fermi surface reconstruction. To get $J_j(-t)$, the two states mixed are the states after the cyclotron motion of a quasiparticle on the FS without current vertex correction (which is different to the state obtained by evolving a quasiparticle with the cyclotron motion around a FS reconstructed by the CVC).

For simplicity in this work, we don't consider the factor $1/(1-\varepsilon_{\mathbf{k}}^2)$ in the computation of $\boldsymbol{J}_{\mathbf{k}}$. The factor $\varepsilon_{\mathbf{k}}$ is taken as a gaussian Gaussian with a fixed breadth and height, using the expression:

$$\varepsilon(\varphi) = \frac{\exp(\varphi - \varphi_n)^2}{2\sigma^2} \tag{42}$$

where φ is the angle of the vector $\mathbf{k}$ and φ_n is the angle of the n-th hot spot. The width of the gaussian σ is at first chosen to be a constant between 0.07 and 0.12. It is somewhat arbitrary in the absence of data allowing us to determine quantitatively the AFM correlation length ξ_{AFM}.

1.3.3 Temperature dependance of the CVC

In this work, we also introduce a temperature-dependance of the CVC. We introduce it mainly in order to fit the experimental values of the Hall number R_H (Section 2.3). We work directly on the quantity $\varepsilon_{\mathbf{k}}$ defining a T-dependance making the height and the width of the gaussian variating as :

$$\varepsilon(\varphi, T) = \alpha(T) \frac{\exp(\varphi - \varphi_n)^2}{2(\sigma\alpha(T))^2} \qquad \text{with} \qquad \alpha(T) = \frac{1}{1 + \exp\left(\frac{T - T_0}{\theta}\right)} \tag{43}$$

T_0 and θ are again arbitrary. We fit them in this work to values likely to fit kinetic quantities such as the Hall number or the resistivity. We often have $\theta = 0.06$ and $T_0 = 10$

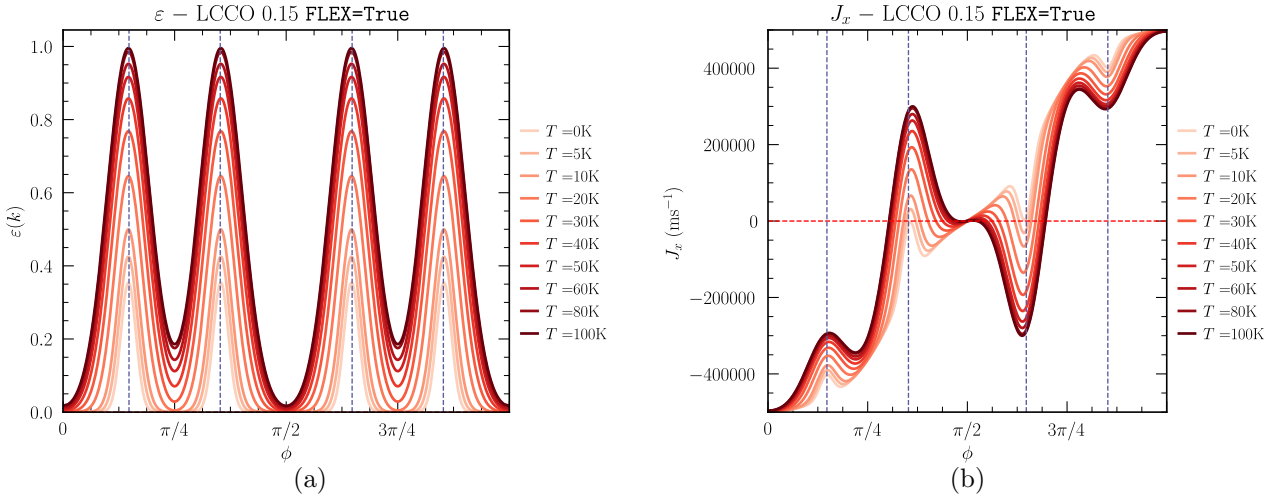


Figure 8 — Temperature variation of the CVC in (a) the intensity of the CVC ε mediated by the AFM correlation length. As T increases the peak of the gaussian gets closer to one and the with of the gaussian increases. (b) Variation of the corrected velocity (or current) J_x. Peaks at the hot spots are increase with T. These peaks allows the product $\ell_x\, \partial \ell_y / \partial \varphi$ to add negative contribution to σ_{xy}

1.4 Scattering rate

To model the scattering rate, we take the model described in [5] by Duffy et al. For the scattering rate, we use the hot spots defined by the AFM fluctuations, as seen in Figure 4. We also add a quasi-isotropic scattering rate to account for the electron-electron scattering. The scattering rate they determined with the minimal ingredients is :

$$\tau^{-1}(T,\varphi) = \overbrace{\tau_{\text{imp}}^{-1} + \tau_{\text{HS}(\varphi)}^{-1}}^{T\text{-independant}} + \overbrace{(g(x)\alpha_1 T + \alpha_2 T^2)\sin^{\nu}(2[\varphi \pm (\pi/4 - \text{HS})])}^{T\text{-dependant}} \tag{44}$$

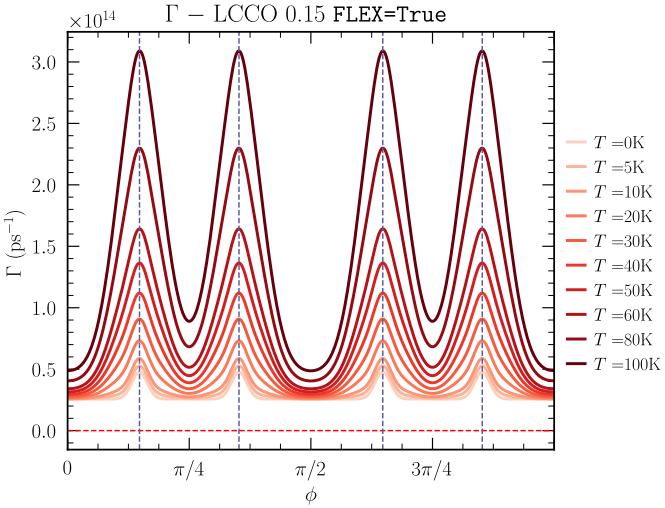
Due to small variations in v_F (inferior to 10%), the anisotropy in the scattering rate is solely contained in the hot spots scattering. This anisotropy is present in both the elastic term and the inelastic term. In the inelastic term, the factor $\sin^{\nu}(2[\varphi \pm (\pi/4 - \text{HS})])$ transcribes a strongly-peaked anisotropy at the hotspots. ν is somewhat arbitrary, in this work we set $\nu = 12$ to account for both strong peaks and a four-fold symmetry of the Fermi surface. $\tau_{\text{HS}}^{-1}(\varphi)$ in the elastic term are gaussians with width $\sigma = 0.05$ centered at the hot spots. These add strongly localised anisotropy that allows the magnetoresistance to be linear at high field, even at very low temperature.

$$\tau_{\text{e-e}}^{-1} = \beta_1 (1 + 0.4 \sin(2\varphi)^2)\, T^2 \tag{45}$$

In order to propose a consistent fit with experimental data, we also add a doping-dependance to the impurity scattering rate: $\tau_{\text{imp}}^{-1} = \tau_0^{-1} \tau_{\text{imp}}^{-1}(x)$. $\tau_{\text{HS}(\varphi)}$ is a gaussian peaked at the hot spots with height α_{HS} and width σ_{HS}

We use in the simulations presented in this work the following scattering rate $\Gamma := \tau^{-1}$:

$$\begin{aligned}\Gamma(T,\varphi) &= \tau_{\text{imp}}^{-1} + \tau_{\text{HS}(\varphi)}^{-1} \\ &+(g(x)\alpha_1 T + \alpha_2 T^2)\sin^{\nu}\left(2\left[\varphi \pm \left(\frac{\pi}{4} - \text{HS}\right)\right]\right) \\ &+\beta_1(1+0.4\sin(2\varphi)^2)\,T^2\end{aligned} \tag{46}$$



(a)

x	$g(x)$	$\tau_{\text{imp}}^{-1}(x)$	$\sigma(x)$
0.14	1.9	1.1	0.16
0.15	1.5	1.2	0.13
0.16	0.5	1.3	0.11
0.18	0.2	1.4	0.09

τ_0^{-1}	τ_{HS}^{-1}	α_1	α_2	β_1
27	$\alpha_{\text{HS}} = 20$ $\sigma_{\text{HS}} = 0.05$	0.75	0.0112	0.0022

(b)

Figure 9 — (a) Scattering rate Γ evolution $\Gamma(\varphi)$ for temperatures between 0 and 100K and (b) Table with the parameters used to compute the scattering rate

2 Results

Magnetoresistance (MR) measurements are typically helpful to understand the origins of normal scattering states. According to the Boltzmann Transport theory, the fractional magnetoresistance is independent of the carrier density and its strength is determined by the magnitude of τ^{-1}, extracted from the mean free path ℓ and its variations around the Fermi surface. MR measurements usually shed light on the pairing mechanisms of electrons *i.e.* the correlation between the scattering strength and T_C. Previous angle-dependant interlayer MR studies (*i.e.* $I \parallel c$) had some success in isolating scattering rate in two families of hole-doped cuprates and corresponding attemps of modeling $\tau^{-1}(\varphi, T)$ [2].

The physics of these materials has yet to be fully grasped. The Fermi Liquid Theory fails to explain anomalous transport phenomenons like the strong H-linearity of the magnetoresistance. Some previous have been able to reproduce the H-linearity of Nd-LSCO [2] thanks to a strong anisotropy in the scattering rate, however modelling of the behaviour of other hole-doped cuprate with the Boltmann Transport Equation has failed [8]. A breakthrough in the understanding of the properties of electron-doped cuprates has been made by Duffy et al. [5]. This recent study on LCCO has been able to offer a model of the scattering rate that, within the Boltzmann framework, leads to computations consistent with experimental values.

The first part of this project was to reproduce the results obtained by Duffy et al. [5] with SCTIF simulations, which core was coded by A. Cuoghi [8]. We were then able to implement the current vertex corrections in this framework, in order to reproduce anomalous behaviour of the Hall effect in LCCO. Previously, the Hall effect couldn't be fitted within in the Boltzmann picture without CVC. We examine in this work a semi-quantitative fitting of the different resistivity quantity (MR, $\rho_{xx}(T)$, R_H). Adding to the CVC, we also study the effect of T-dependance and doping-dependance of the fitting parameters of resistivity values.

The aim of this work is to first of all compute all results using previous computations [8] and add a quantitative approach to qualitative results of the computation of MR and Hall Effect in LCCO done in [5].

2.1 Previous study of electron-doped cuprates

A comprehensive study of the MR and in-plane resistivity of LCCO overdoped region ($x = 0.116$ to $x = 0.188$) has been done by Duffy et al. [5]. They were able to offer a model that reproduces the main features of the transport properties of LCCO.

The temperature and doping dependance of the resistivity can be captured within a semi-classical Boltzmann transport picture. This is allowed by the introduction of so-called "hotspots" in the fermi surface. These hotspots located either at the nodes or the antinodes of the Fermi surface (Figure 10) located either at the antiferromagnetic Brillouin zone or at the boundaries of the first Brillouin zone. The nodes hotspots account for antiferromagnetic, *i.e.* spin, fluctuations in LCCO and the antinodes for charge fluctuations. These strong scattering centres act to impede cyclotron motion [16] and preserve the H-linearity of the MR.

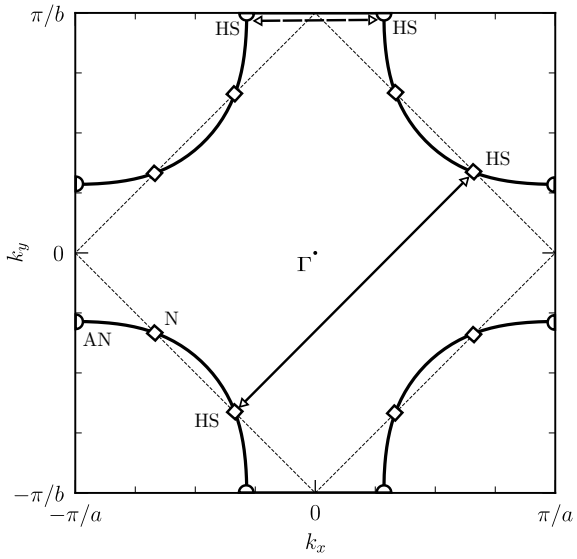


Figure 10 — Fermi surface of LCCO ($x = 0.15$). The long diagonal arrow represents the coupling between two hot spots by a scattering vector $Q = (\pi, \pi)$. The dotted lines represent the antiferromagnetic Brillouin zone (AFBZ), half the size of the Brillouin zone (BZ) since the unitary cell in real space doubles in size due to antiferromagentic structure. Also marked are the nodes of the supercoductor at the intersection of the AFBZ boundary and the FS and the antinodes at the intersection between the BZ boundary and the FS

The key properties of the magnetoresistance this study aims to model are the following:

1. The zero-field resistivity below 60K can be described accurately by the form $\rho_0 + A_1 T + A_2 T^2$. With increasing x, A_1 falls while A_2 remains constant, suggesting that only the T-linear term is x-dependent [17] (see Figure 1).
2. The anisotropy in ℓ must lie predominantly in the inelastic (T-dependent) factors since the magnitude of the MR falls with decreasing T [5]
3. The strong H-linear MR at high field strengths is a key feature that mainly failed to be captured within the Boltzmann framework in hole-doped cuprates studies. It is here accounted thanks to a strong anisotropy in the hotspots.

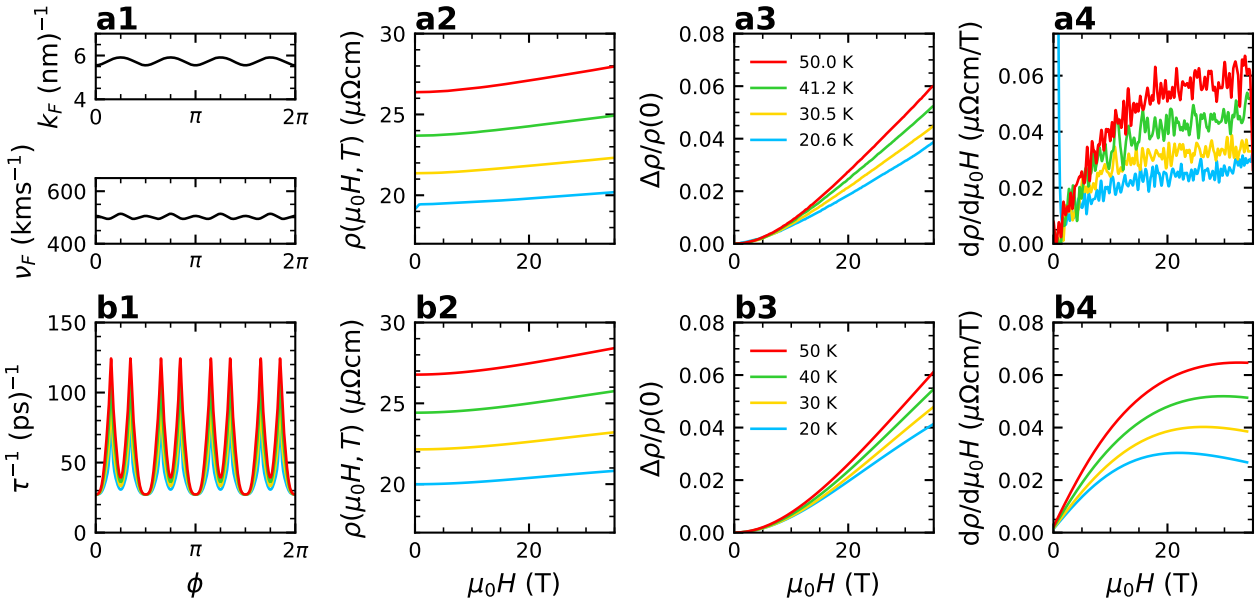


Figure 11 — a1) Variation of the FS (upper panel) and Fermi velocity ν_F obtained from the tight-binding parameterisation. The anisotropy in the velocity is very weak (less than 10%) so that we consider the anisotropy lying mainly in the scattering rate **a2-a4)** experimental data of magnetoresistance of LCCO at low temperatures. We obeserve a strong linear-in-field component, **b1)** Scattering rate as described in eq. 44 and **b2-b4)** simulations of the resistivity and the fractional MR and its derivative. The temperature and field behaviour have been successfully reproduced by the model. Adapted from [5]

Their results Figure 11 show succes in the modelisation of the magnetoresistance with the hot spots model. The derivative of the magnetoresistance $\mathrm{d}\rho_{xx}/\mathrm{d}B$ show a robust H-linear component dominant at low T. Ther were also able to fit parameters to have consistent orders of magnitude for ρ_{xx} and MR with experimental values.

A major aspect of the physics of cuprates that still fails to be captured by this model is the behaviour of the Hall coefficient (ρ_{xy}/B) in both field and temperature. A qualitative study of current vertex correction in their work enabled them to have encouraging results. The aim of this work is to do a quantitative study of the effect of the CVC on the different aspects of the resistivity in the Boltzmann Transport picture, and to fit a model as close as possible to experimental results.

2.2 First implementation of the CVC

Without current vertex correction, we were able to reproduce the results obtained by Duffy et al. [5] (see Figure 11) for $x = 0.15$. The main features of the MR are reproduced (linear in $\mathbf{B}$, magnitude goes down with temperature). As shown in Figure 12c, the Hall effect is constant at low temperature and goes slightly up towards 100K. It is an expected behaviour for computations done without the current vertex corrections.

We add the current vertex corrections to the computation, with constant intensity versus T. We have indeed a reduction of the Hall number. It is reduced by 10% for a width of the intensity of the CVC $\sigma = 0.1$, peak of $\varepsilon = 0.9$, and high-field values fit experimental results as the width increases $\sigma = 0.14$, peak of $\varepsilon = 1$.

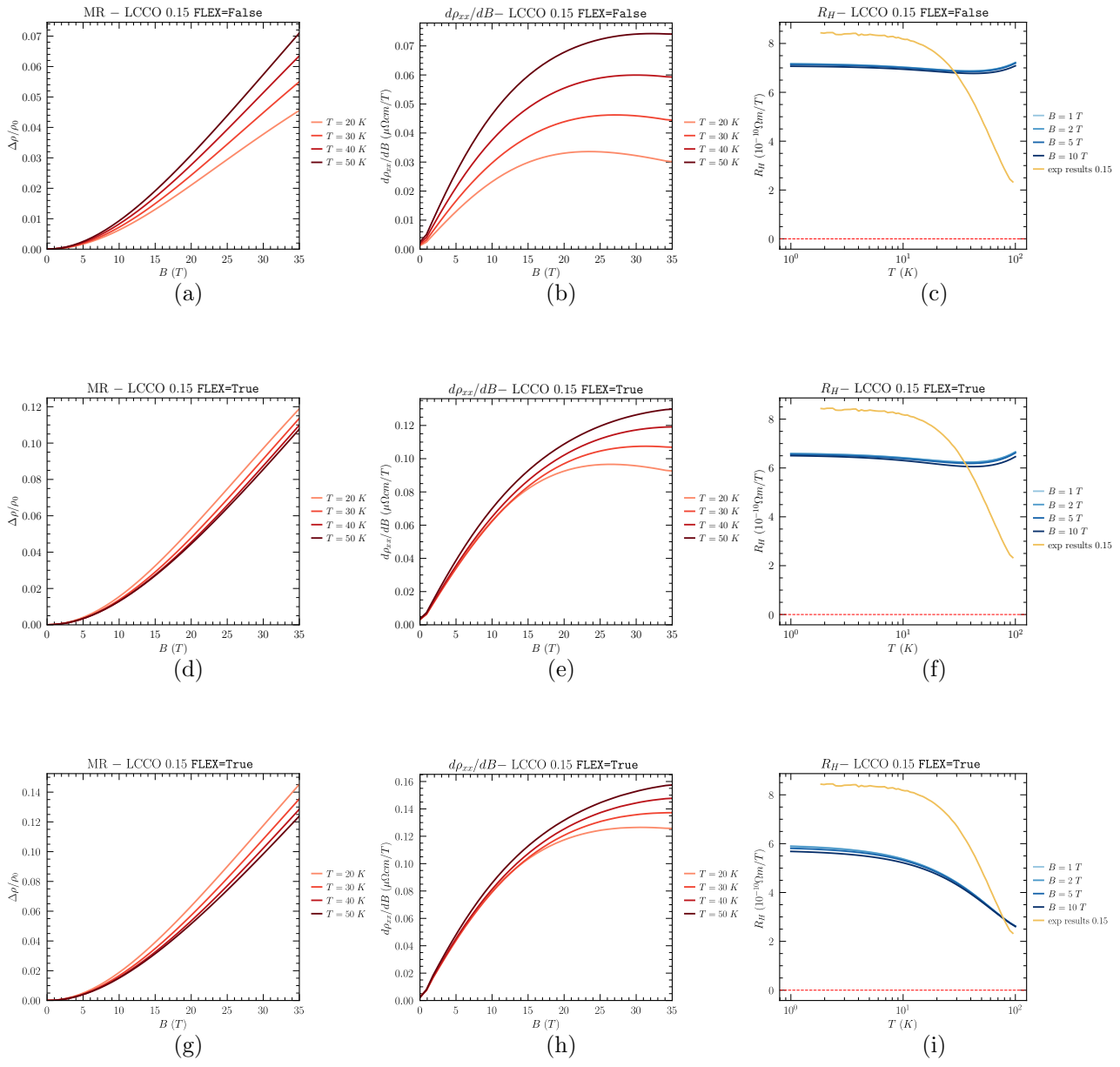


Figure 12 — Computation of MR, dMR/dB and R_H. (a – c) Computation without current vertex correction, the MR is consistent with previous model, however the Hall effect doesn't behave as expected. (d – f) Computations with low CVC: $\sigma = 0.1$ and the height of the peaks in ε is 0.9. MR is slightly increased and R_H is decreased. (g – i) Computations with intense CVC: $\sigma = 0.14$ and height of ε goes up to 1. The Hall effect tends towards the experimental values at high temperature, however the Drude value is decreased and the magnitude in the MR seems to merge in only one curve

As we add CVC to the model, the T-dependance of the magnetoresistance is modified. The **B**-linear behaviour stays predominant but the magnitude of MR decreases T increases. Moreover, the MR is slightly enhanced because of the CVC. However, our main concern with this model is the value of R_H at 0K. It should not variate and stay close to the Drude value (see experimental curves). It tends towards 0 as we increase the CVC. A solution to this problem has been to introduce T-dependance in the intensity of the CVC, a way of *turning on* the fluctuations as T increases.

2.3 Temperature dependance of CVCs

As presented in Section 1.3.3, we introduce a progressive increase of the intensity of the current vertex correction.

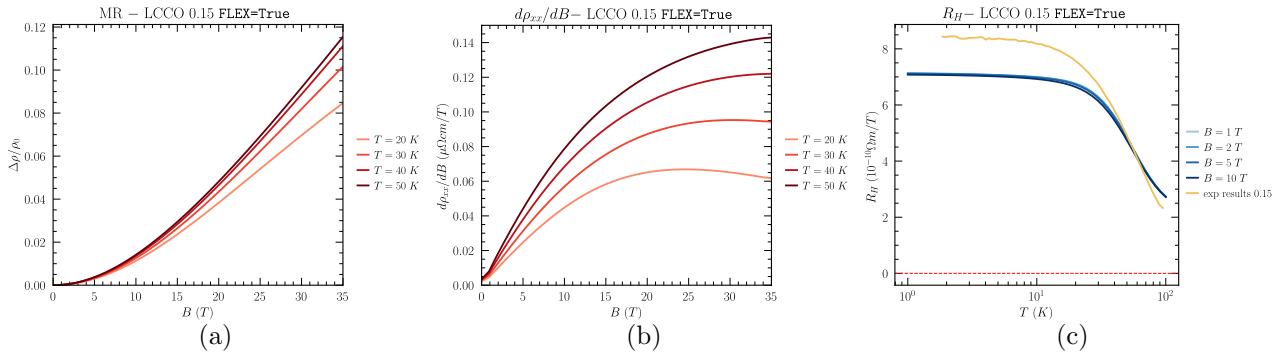


Figure 13 — Current vertex correction's intensity increases the temperature. The Hall number and the MR correspond better to experimental results or previous modelisations compared to a constant intensity of the CVC in temperature.

This modification in the model helps the Hall number to better fit the experimental values. Moreover, magnetoresistance is less impacted by the current vertex corrections and its magnitude increases with T as shown in [5].

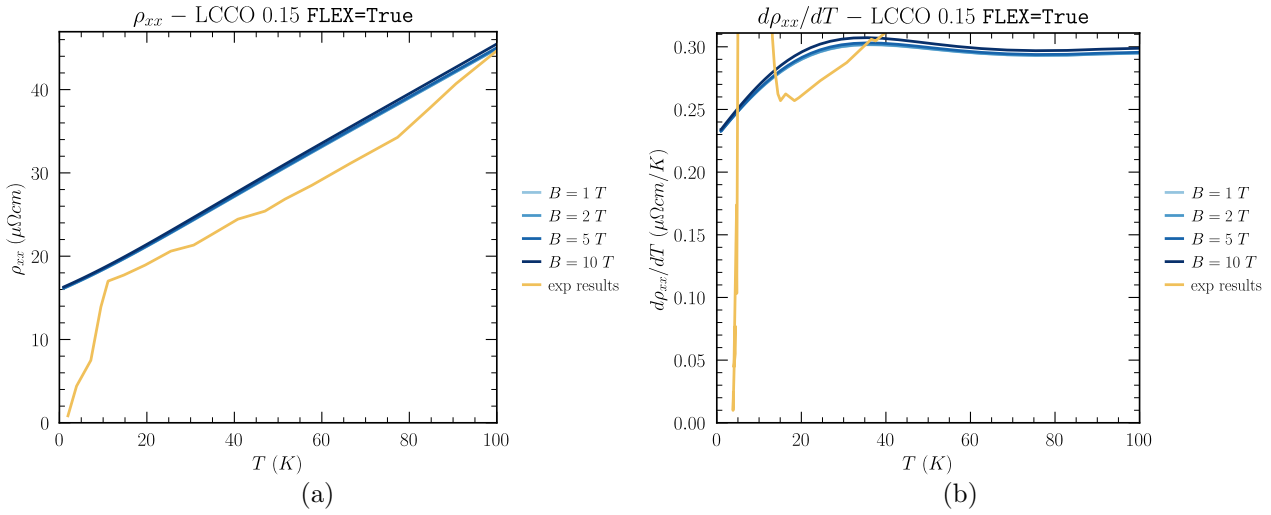


Figure 14 — Magnetoresistance and resistivity simumations. The resistivity of LCCO is "bumped" as shown in (b). The derivative switches between two values due to T-dependant current vertex corrections.

However a problem remains in the longitudinal resistivity. The resistivity is mainly quadratic in LCCO and goes from one behaviour (no CVC) to the other (full CVC) as the temperature increases. It causes a "bump" to appear in the derivative of the resistivity as this quantity switches between

two behaviours (Figure 14). In order to compensate for this bump, we add to the scattering rate the contribution of electron-electron scattering (eq. 45).

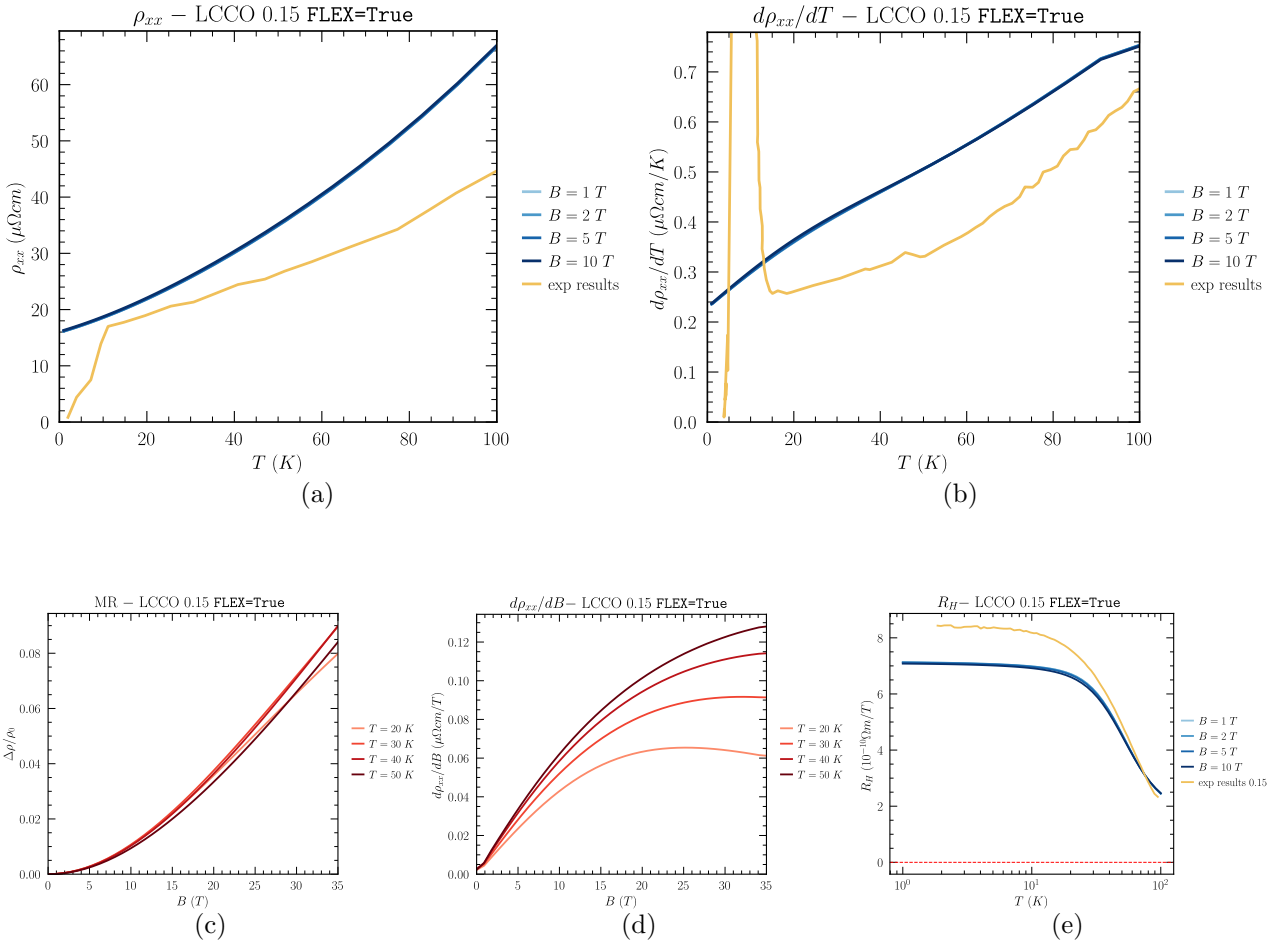
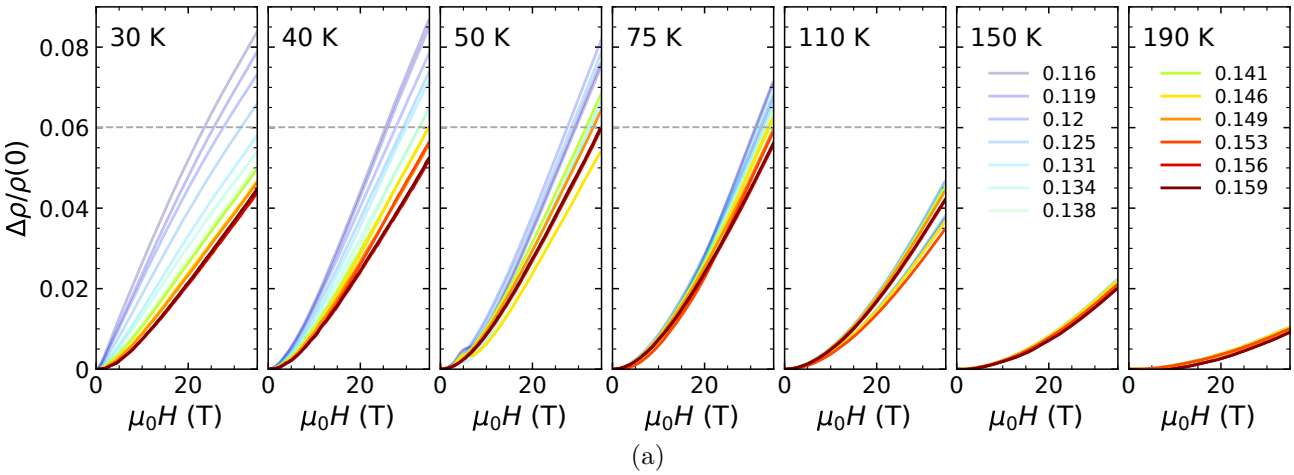
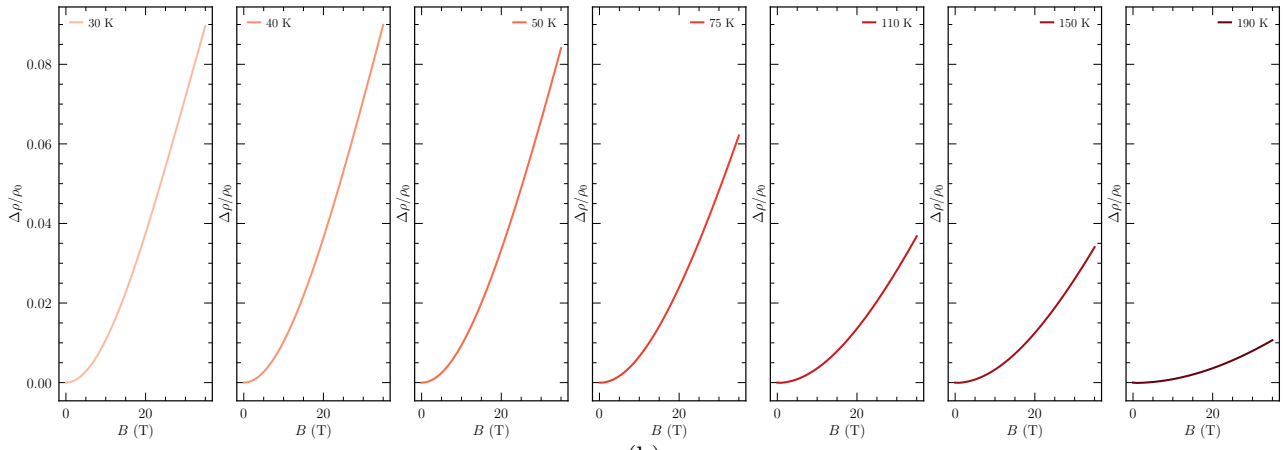


Figure 15 — Magnetoresistance and resistivity simumations. (a – b) The "bump" is reduced, almost inexistant, in resistivity as well as in the derivative of the resistivity. (c – e) MR and R_H stay consistent with experimental values and previous modelisations even if the MR decreases too quickly with T, as it is measured to increase up to 70K

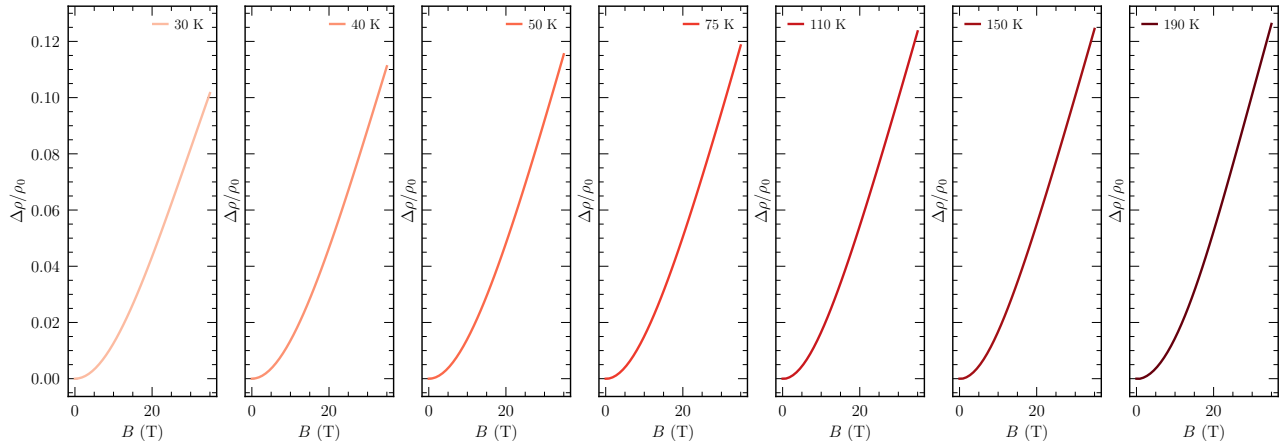
This fixes the bump and makes the evolution of MR more consistent with experimental values compared to the MR without electron-electron scattering. The MR evolution of the temperature was not going down with temperature even at high temperatures Figure 16c. However, even if the behaviour in the MR with electron scattering does not merges perfectly with experimental data, its behaviour is rather consistent. It further indicates that the electron-electron scattering is needed in this model.



(a)



(b)



(c)

Figure 16 — (a) Experimental data of MR for different dopings of LCCO and several temperatures. Adapted from [5]. (b) Temperature evolution of the MR for LCCO 0.15, the MR increases slightly between 30 and 40K, and then goes down in temperature as shown in (a). (c) MR computed without electron-electron term in the scattering rate, the MR keeps increasing with temperature, contrary to the experimental values.

2.4 Doping study of LCCO

Fitting the doping behaviour of the resistivity is crucial in superconductors. Doping and T_C are indeed linked, if the doping behaviour of LCCO is reproducible, it enforces the possible link between superconductivity and spin fluctuations.

Without doping dependance of the parameters, the doping behaviour of the resistivity is contradictory with the experimental quantities. However, when adding doping parameters in the scattering rate, we were able to fit it better than previously. The fitting realised in this work (Figure 17) still needs to be optimized, but with the present modelisation, we could not obtain a better result. All doping parameters are listed in (Figure 9). The doping parameter $g(x)$ decreases with doping [5]. The impurity scattering doping dependance is increasing with doping. The low-T resistivity, *i.e.* above T_C is doping-dependant, and this offset is increasing with temperature [5].

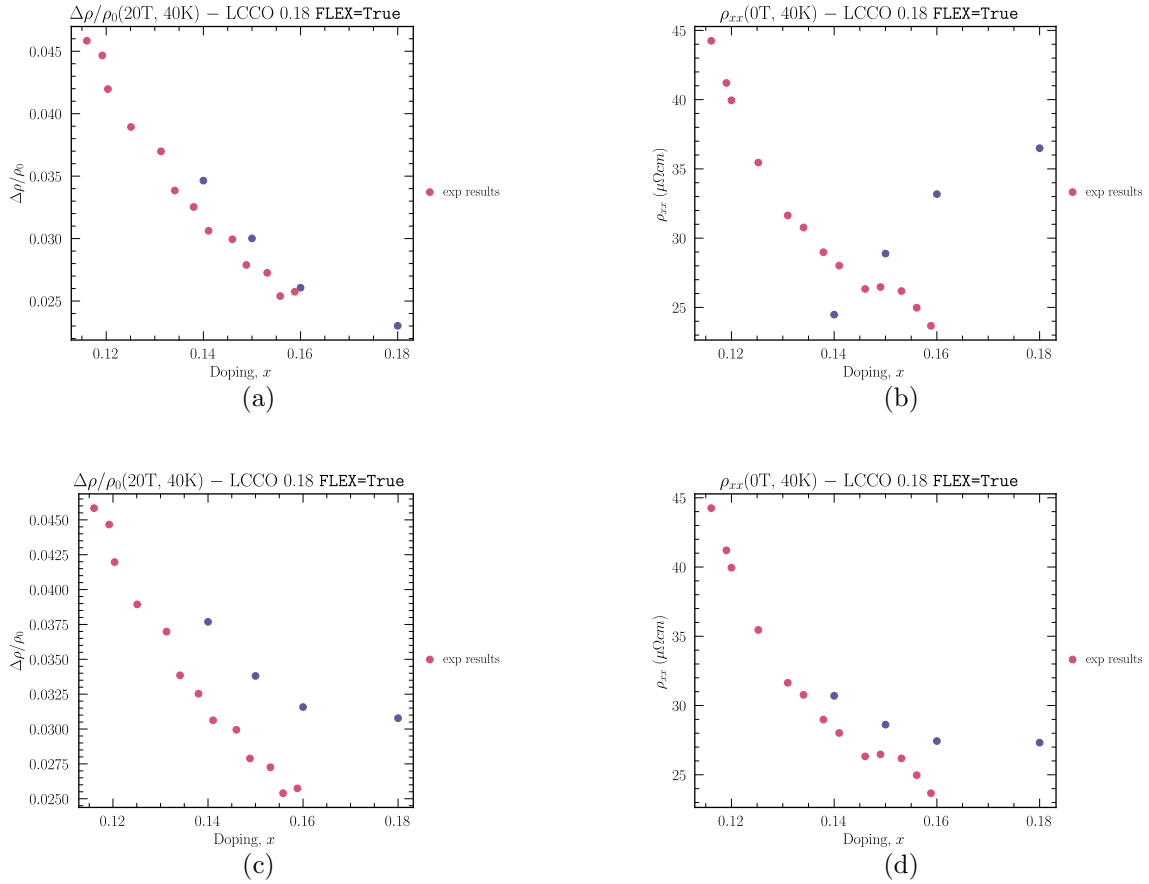
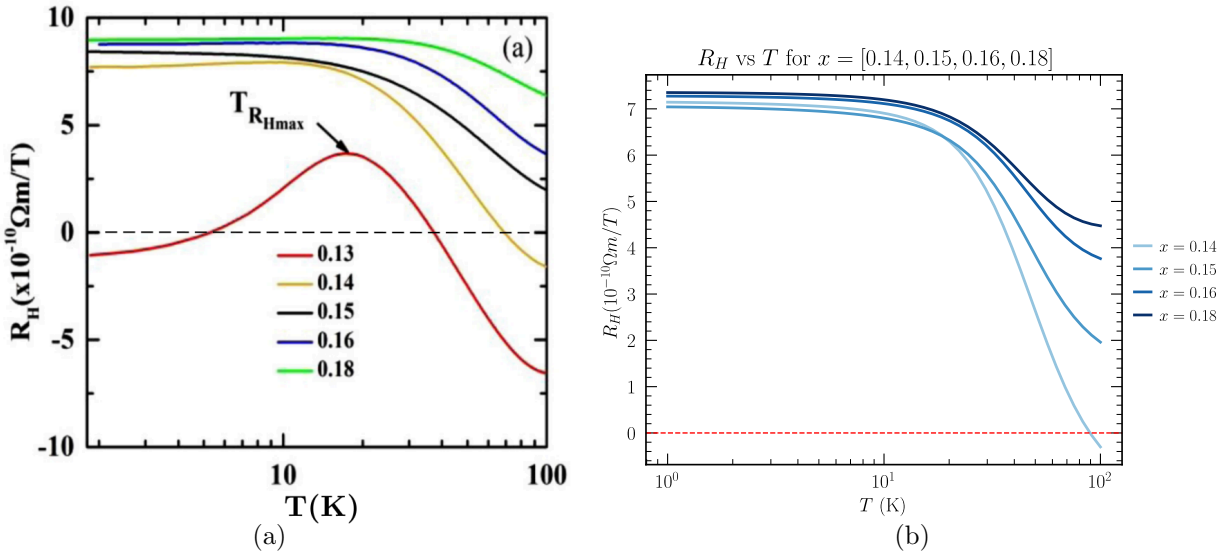


Figure 17 — (a) Fractional MR at 20T and 40K *vs* doping without doping dependant impurity scattering fits very well experimental data from [5], however (b) the resistivity at 0T and 40K *vs* doping without impurity scattering is increasing with doping instead of descreasing. Experimental data from [5]. (c) Fractional MR at 20T and 40K *vs* doping with doping dependant impurity scattering. The doping parametrisation is the one described in Figure 9 (d) the resistivity at 0T and 40K vs. doping is now increasing with doping instead of decreasing

The doping evolution of the Hall number is encouraging regarding the experimental values (Figure 18). The parametrisation of the tight binding makes the simulations tend toward the exper-

imental values (the variation in R_H increases as the doping decreases). To fit the experimental curves of the Hall effect, we added a doping dependance in the correlation length of the CVC (see Figure 9).



(a) (b)

Figure 18 — Hall number R_H vs. T for different dopings (a) Experimental data reproduced from [13] (b) Simulations computed with above parametrisation of CVC, for dopings above x_{FSR} 0.14, 0.15, 0.16 and 0.16. The value of the Hall number at 0K is lower than the experimental (Drude) value in the simulation. However, the temperature and doping behaviour has been succesfully reproduced.

2.5 Charge fluctuations

Another type of fluctuations that can be considered in the CVC picture is the charge fluctuations. If the scattering vector Q is modified and connects the antinodes (dotted line vectors in Figure 10), the CVC can be mediated through charge fluctuations. In the study done by Duffy et al., they find that the charge fluctuation can lead to H-linear MR. In their semi-quantative computations of the Hall number, they show that it leads to increase of the Hall number with temperature, contrary to experimental results. Indeed, as the scattering rate is the strongest in antinodal regions, the mean free path is very small in regions where the circulation of ℓ is opposed to the circulation of φ. We have in this work computed more detailed simulations that lead to the same conclusion. The Hall number R_H is considerabely enhanced with charge-fluctuations mediated CVC (Figure 19) even if the magnetoresistance is consistent with experimental results.

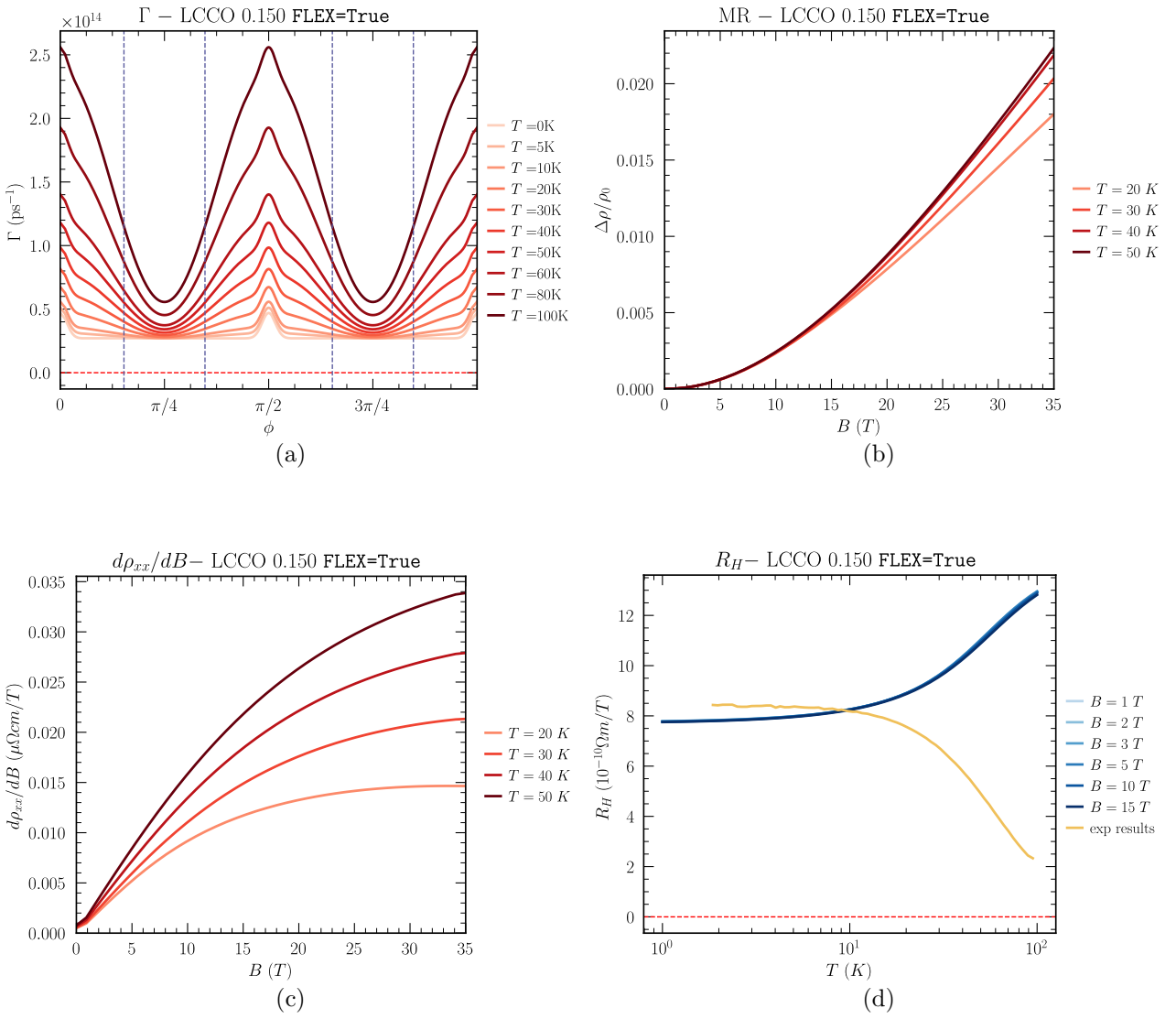


Figure 19 — Charge fluctuations transport simulations (a) Scattering rate with peaked anisotropy at the charge hot-spots located at the antinodes. (b – c) Magnetoresistance is consistent with experimental data, even if the order of magnitude with the scattering rate parametrisation is halved compared to experimental results. (d) Due to peaked scattering rate and CVC in the antinodal region, the Hall number is strongly increased with CVC as temperature increases. This goes against experimental results.

2.6 Hole-doped cuprates

The previous study was done on LCCO, an electron-doped cuprate. The implementation of spin fluctuations in the model gave encouraging results to analyse the behaviour of electron-doped cuprates in the Boltzmann picture. The Fermi liquid theory seems to hold for electron-doped cuprates normal regime. In this section, we report the attempts to fit the same model on hole-doped cuprates. We

implement the hot spot model of the scattering rate (strongly peaked at the intersection between the Fermi surface and the AFBZ boundary) for hole-doped cuprates.

Contrary to only commensurate spin fluctuations in electron-doped cuprates [18], fluctuations are not commensurate for hole-doped cuprates [19]. Each scattering vector $\mathbf{Q} = (\pm\pi, \pm\pi)$ becomes four incommensurate scattering vectors: $(\pi + \delta, \pi + \delta)$, $(\pi - \delta, \pi + \delta)$ and their opposites. δ ranges from 0.05 to 0.15 in LSCO [19]. We use an "effective" AFBZ to compute the position of the hot spots and the states they are mixed with because of current vertex corrections.

The effective result in the scattering rate and in the computation of the velocity is an effect of broadened hot spots as there is a cluster of hot spots (see Figure 20a)

The hot spot modelisation in the scattering rate fails for the hole-doped cuprates. We were unable to reproduce the H-linearity of the magnetoresistance. Even if the current vertex corrections enhance the hall effect with temperature, we were not able to reproduce MR properties in this approach.

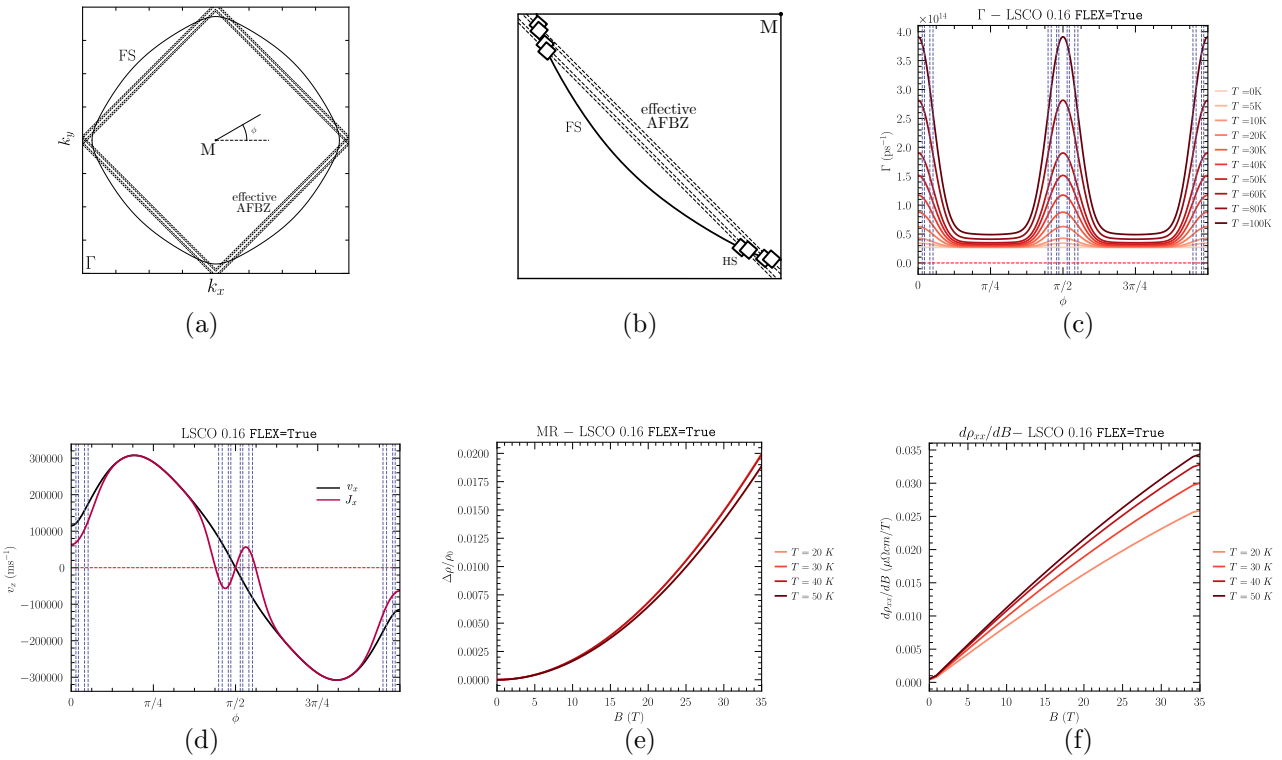


Figure 20 — (a–b)Fermi surface of LSCO centered on the M-point with incommensurate "effective" AFBZ that determines the location of the different hot spots. Since there are four incommensurate scattering vectors, there is four times more hot spots than in the commensurate case. Each hot spot scatters only with one other point on the Fermi surface (c) Scattering rate vs. φ. The hot spots (vertical dotted lines) are broadened due to incommensurate scattering. (d) Fermi velocity and its counterpart taking into account current vertex corrections. (e – f) Transport simulations of magnetoresistance and its derivative with CVC activated. The MR is clearly quadratic in field, which is not consistent with experimental results.

3 Discussion and conclusion

In this work, we have been able to reproduce some of the key features of the resistivity in electron-doped cuprates. To do so, we compute the resistivity of LCCO with the Shockley-Chambers tube integral, a formalism developped within the Boltzmann transport framework. To obtain consistent magnetoresistance and Hall number simulations with experimental results, we take into account the notion of "hot spots" mediated by spin fluctuations. We also implement further effect of these fluctuations through the current vertex corrections. Since the main goal of this work is to have simulations as close as possible to experimental results, a key question remaining is the physical meaning and the relevance of the modelisation.

For current vertex correction, we modify the velocity given the formula in eq. 39. A key parameter is the strengths of the CVC $\varepsilon(\varphi)$, modelised throughout this work by a gaussian. However in reality, it is a more complex element. It can be approximated by $\varepsilon(\varphi) \approx \left(1 - c\xi_{\mathrm{AFM}}^2\right) < 1$, where c is a constant and ξ_{AFM} is the antiferromagnetic correlation length [12]. In this work we don't the correlation length into account at all. It still is a key parameter of the physics of cuprates, especially in the temperature dependance. At low temperatures, in our modelisation the strength of the current vertex correction increases up to almost one, and the width of the correction is spreading up to the next hot spots. Physically, the increase in temperature can cause this type of fluctuations to gain importance, it is justified by the relation [20]

$$\xi_{\mathrm{AFM}} \propto \frac{1}{aT + b} \tag{47}$$

Which gives an epsilon getting closer to one as temperature increases. The temperature study in this work is however quantitative, and made to fit at best all the resistivity parameters, without any study actually tied to experimental results.

While the CVC allows us to model the temperature-dependance of the Hall effect in LCCO, we also note that the order of magnitude of the width of the CVC is arbitrary. The width σ goes up to 0.15, which is a pretty important broadness of the hotspot. The point between two hotspot adds a finite contribution to the velocity as $\varepsilon(\varphi = \pi/4) \neq 0$. This contribution can seem too high to be the only effect bringing the Hall effect to decrease with temperature enough to match experimental data.

Although the doping fitting has been made consistently with physical values of impurity scattering, the doping fitting with current vertex correction is not optimized yet in this work. Without CVC, the doping fitting done in previous study [5] is really close to experimental values. However in our simulations, as doping increases we struggle to define parameters decreasing enough to make both the resistivity and the magnetoresistance decrease with doping. This means that the doping dependance of the model has to be investigated further. The only doping study that we do here is on four samples $x = 0.14$, 0.15, 0.16 and 0.18 because these are then only doping values we have Hall number measurements data for. However, a more comprehensive doping study can be done on more doping samples.

The electron-doped study developped from experimental results and computations done by Duffy et al. is not exhaustive but frames the main experimental features of the material. The hole-doped study in this work is however very crude and has to be taken further. Here we simply transpose the scattering rate model we used with LCCO on hole-doped cuprates, taking into account the incommensurates hotspots. It doesn't give relevant magnetoresistance results at all. However, we haven't computed other scattering model that were previously proven to give H-linear magnetoresistance [2,3] with the presence of current vertex corrections. The present work is then far from sufficient to

draw conclusion on the Fermi liquid theory and Boltzmann transport model in hole-doped cuprates.

Essentially in this work, after developping the appropriate theoritical framework to apprehend transport properties of electron-doped cuprates, we have been able to push further features of a previous scattering rate model to fit experimental data. We have also implemented the current vertex correction in the Boltzmann transport framework. We have considered temperature as well as doping dependance of the CVC to fit at best LCCO transport data. Even if this work if far from complete and does not allow us to draw any conclusion on hole-doped cuprates, it provides interesting results on LCCO and the role of spin fluctuations in the mediation of the superconductivity.

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